Infectious Diseases II

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Learning Objectives

1. Compose a plan to incorporate quality metrics (e.g., prevention of catheter-associated urinary tract infections and catheter-related bloodstream infections) into pre- and postsurgical care.
2. Identify key members of, common strategies, and tools (including biomarkers and rapid diagnostic tests) used by an antimicrobial stewardship team.
3. Provide empiric antibiotic therapy recommendations for critically ill patients with community-acquired or health care–associated meningitis.
4. Evaluate therapeutic options for the treatment of multidrug-resistant pathogens in the intensive care unit (ICU).
5. Devise an optimal treatment plan for critically ill immunocompromised patients with infectious diseases.

Abbreviations in This Chapter

ACS NSQIP	American College of Surgeons National Surgical Quality Improvement Program
AST	Antimicrobial susceptibility testing
CAUTI	Catheter-associated urinary tract infection
CDC	Centers for Disease Control and Prevention
CLSI	Clinical & Laboratory Standards Institute
CMS	Centers for Medicare & Medicaid Services
CNS	Central nervous system
CoNS	Coagulase-negative *Staphylococci*
CRE	Carbapenem-resistant Enterobacterales
CSF	Cerebrospinal fluid
ESBL	Extended-spectrum β-lactamase
HAART	Highly active antiretroviral therapy
IDSA	Infectious Diseases Society of America
ICU	Intensive care unit
KPC	*Klebsiella pneumoniae* carbapenemase
LOS	Length of stay
MALDI-TOF	Matrix-assisted laser desorption-ionization/time of flight
MIC	Minimum inhibitory concentration
MRSA	Methicillin-resistant *Staphylococcus aureus*
MSSA	Methicillin-susceptible *Staphylococcus aureus*
OI	Opportunistic infection
PCR	Polymerase chain reaction
PCT	Procalcitonin
PD	Pharmacodynamic(s)
PK	Pharmacokinetic(s)
PNA FISH	Peptide nucleic acid fluorescence in situ hybridization
SCIP	Surgical Care Improvement Project
SOT	Solid organ transplantation
SSI	Surgical site infection
UTI	Urinary tract infection
WHO	World Health Organization

Self-Assessment Questions

Answers and explanations to these questions can be found at the end of this chapter.

1. A 56-year-old man (weight 140 kg) is scheduled to have elective coronary artery bypass surgery with aortic valve bioprosthetic replacement. The patient has a medical history of diabetes and hypertension. He has no drug allergies and has a calculated CrCl of 90 mL/minute/1.73 m^2. It is anticipated that the patient will be admitted to the cardiovascular surgery ICU after the surgery with two chest tubes placed for drainage. Which is the most appropriate antimicrobial prophylactic regimen for the patient?
   A. Cefazolin 2 g once within 2 hours before incision and administered every 8 hours until chest tubes are removed.
   B. Vancomycin 2 g once within 2 hours before incision and administered every 12 hours until 48 hours after surgery.
   C. Cefazolin 3 g once within 1 hour before incision, re-dosed if surgery lasts more than 4 hours, and continued for 48 hours after surgery.
   D. Cefazolin 2 g once within 1 hour before surgery, re-dosed if surgery lasts more than 4 hours, and continued for 48 hours after surgery.

2. Which statement best describes antimicrobial stewardship in the ICU?
   A. Formal antimicrobial stewardship team membership should be reserved for clinical pharmacists with infectious diseases training.

B. Antibiotic cycling is helpful in minimizing antimicrobial resistance.
C. Daily clinical activities of a critical care pharmacist may constitute an antimicrobial stewardship effort.
D. Antimicrobial stewardship should not be instituted in critically ill patients because restriction of antimicrobial choices may worsen outcomes.

3. A 33-year-old woman with a history of hydrocephalus since early childhood requiring placement of an internal cerebrospinal fluid (CSF) shunt, presents with a temperature of 102.8°F (39.3°C), an altered mental status, and a white blood cell count of 19 x 10^3 cells/mm^3. After a computed tomography scan reveals no hydrocephalus or midline shift, a lumbar puncture is performed. Initial cell count shows elevated CSF white blood cells with a high proportion of neutrophils and low glucose. Neurosurgery is consulted for a presumed CSF shunt infection. Which empiric systemic antibiotic regimen is most appropriate?
   A. Ceftriaxone and ampicillin.
   B. Ceftriaxone and vancomycin.
   C. Piperacillin/tazobactam and tobramycin.
   D. Cefepime and vancomycin.

4. A 75-year-old man is admitted to the medical ICU from the emergency department with septic shock. He is fluid resuscitated and administered broad-spectrum antibiotics with piperacillin/tazobactam and vancomycin. On day 2 of therapy, the patient remains hemodynamically unstable, requiring norepinephrine. The patient's blood culture is positive, according to the institution's microbiology laboratory, which recently implemented rapid diagnostic tests using nanoparticle microarray technology, and the microarray results show the presence of *Escherichia coli* with the CTX-M resistance gene. Which intervention regarding the patient's antimicrobial therapy is most appropriate?
   A. Change piperacillin/tazobactam to cefepime.
   B. Add tobramycin to the current regimen.
   C. Add levofloxacin to the current regimen.
   D. Change piperacillin/tazobactam to meropenem.

5. An 81-year-old man is admitted to the medical ICU with presumed line-associated sepsis. Blood cultures are obtained that grow pan-sensitive *Enterobacter cloacae*. The patient is being treated with intravenous ceftriaxone, and the offending central venous catheter is removed. The patient initially responds, but on day 10 of therapy, he becomes febrile, and blood cultures are re-sent. At that time, the patient was maintained on ceftriaxone because he was hemodynamically stable. Gram stain for the new blood cultures is positive for lactose-positive gram-negative bacilli. Which is the most appropriate action for this patient's antimicrobial management?
   A. Change ceftriaxone to ceftazidime.
   B. Change ceftriaxone to piperacillin/tazobactam.
   C. Change ceftriaxone to meropenem.
   D. Continue ceftriaxone alone.

6. An 86-year-old woman with a history of end-stage renal disease is admitted to the hospital with respiratory distress requiring intubation, fluid resuscitation, and hemodynamic monitoring. Bronchoalveolar lavage cultures show methicillin-resistant *Staphylococcus aureus* (MRSA). The patient is being treated with intravenous vancomycin. On day 4 of therapy, the patient develops fever, leukocytosis, and erythema around the insertion site of her tunneled dialysis catheter. Blood cultures are sent, and the dialysis catheter is removed. Gram stain from blood cultures is significant for gram-positive cocci in pairs and chains. The medical team discontinues vancomycin and approaches you to inquire about treatment options for this patient. Which agent is most appropriate for this patient?
   A. Daptomycin.
   B. Linezolid.
   C. Ceftaroline.
   D. Tigecycline.

7. A 35-year-old man is admitted to the medical ICU with respiratory distress. He has a 3-week history of cough and pleuritic chest pain that has worsened with time. Chest radiography is performed, which shows bilateral infiltrates with ground-glass opacities. The patient is HIV positive and not currently receiving antiretroviral therapy because of nonadherence. The patient requires intubation and is receiving 40% fraction of inspired oxygen

(Fio_2). His relevant laboratory values are as follows: SCr 1.0 mg/dL, $CD4^+$ count 100/mm^3, *lactate dehydrogenase (*LDH) 550 IU/L, partial pressure of arterial oxygen (PaO_2) 80 mm Hg, partial pressure of arterial carbon dioxide ($PaCO_2$) 40 mm Hg, and white blood cell count (WBC) 4 x 10^3 cells/mm^3. The patient has a sulfa allergy and a glucose-6-phosphate dehydrogenase deficiency. In addition to adjunctive corticosteroids, which regimen for presumed *Pneumocystis jiroveci* pneumonia is most appropriate?

A. Trimethoprim/sulfamethoxazole 15–20 mg/kg intravenously, divided every 6 hours.
B. Pentamidine 4 mg/kg intravenously every 24 hours alone.
C. Primaquine 30 mg orally once daily plus clindamycin 600 mg intravenously every 8 hours alone.
D. Atovaquone 750 mg orally every 12 hours.

8. A 50-year-old woman with acute myeloid leukemia is admitted for induction therapy. During the patient's clinical course, she develops respiratory distress, for which she is admitted to the medical ICU. She is found to have neutropenic fever and is empirically treated with meropenem plus gentamicin plus vancomycin. On day 5 of empiric therapy, blood cultures are growing pan-susceptible *E. coli*. The patient's relevant laboratory values and vital signs are as follows: blood pressure 115/70 mm Hg, heart rate 84 beats/minute, temperature 101.2°F (38.4°C), and absolute neutrophil count less than 50 cells/mm^3. Which is the most appropriate antimicrobial regimen for the patient?

A. Continue meropenem, gentamicin, and vancomycin for 14 days.
B. Discontinue all empiric antimicrobials, and initiate ampicillin intravenously to treat until neutrophils recover.
C. Discontinue vancomycin, and continue meropenem and gentamicin until neutrophil recovery.
D. Add voriconazole, and continue all other antimicrobials for 14 days.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
    a. Task A: 1-4
2. Domain 2: Therapeutics and Patient Management
    a. Task A: 1-4
    b. Task B: 8, 12, 13
    c. Task C: 1, 3, 4
3. Domain 3: Professional Practice
    a. Task A: 3, 4
    b. Task B: 1

I. QUALITY IMPROVEMENTS

A. The American College of Surgeons National Surgical Quality Improvement Program (ACS NSQIP) provides general best-practice guidelines. ACS NSQIP also provides a surgical risk calculator that, depending on the type of surgery and the patient's baseline characteristics, can generate estimated risks of complications, which include pneumonia, cardiac complications, surgical site infections (SSIs), urinary tract infections (UTIs), venous thromboembolism, renal failure, discharge to a rehabilitation facility, and death. (This risk calculator can be found at http://riskcalculator.facs.org/.)

B. As of 2016, all Surgical Care Improvement Project (SCIP) measures have been retired because of consistently high compliance rates. SCIP measure reporting is no longer mandatory. However, the practices described within the retired SCIP measures still represent best practices and should be continued. These include administration of an appropriate prophylactic antimicrobial within 1 hour of incision, discontinuation of the prophylactic antibiotic within a specified time, control of post–cardiac surgery serum blood glucose, and removal of the urinary catheter by postoperative day 2.

C. Sepsis Bundle Project (SEP-1): Specifics regarding early broad-spectrum antimicrobial administration in the management of severe sepsis and septic shock can be found in the Shock chapter.

D. Prevention of SSIs
   1. Epidemiology and clinical significance
      a. More than 290,000 SSI cases occur each year.
      b. Mortality rate is 2–12 times higher in patients who experience SSIs than those who do not.
      c. SSIs account for $4–$10 billion in direct costs.
   2. Risk factors for SSIs: Advanced age, obesity, hyperglycemia, dyspnea, hypoxia, smoking, alcoholism, recent radiotherapy, preoperative albumin less than 3.5 mg/dL, total bilirubin greater than 1.0 mg/dL, trauma/shock, transfusion, hypothermia, inadequate skin preparation, abdominal surgery, contaminated procedures, cancer, emergency surgery, staphylococcal colonization, and prolonged procedures.
   3. The Centers for Medicare & Medicaid Services (CMS) no longer provides reimbursement to providers for the treatment of SSIs after cardiac, bariatric, or orthopedic surgical procedures.
   4. Prevention strategies: Pharmacy specific
      a. Preoperative
         i. Control serum blood glucose in patients with diabetes – ACS NSQIP measure
         ii. Administer prophylactic antibiotics within 1 hour before surgery (vancomycin and fluoroquinolones should be administered within 2 hours before surgery because of prolonged infusion times) – ACS NSQIP measure
         iii. Select the appropriate prophylaxis – See Table 1.
            (a) 10%–15% of the U.S. population report an allergy to penicillin antibiotics and 1%–2% to cephalosporin antibiotics; these allergy labels often result in avoidance of cefazolin and use of alternative antibiotics (e.g., vancomycin, clindamycin).
            (b) Retrospective cohort studies have found that patients with a reported penicillin or cephalosporin allergy had a higher odds of developing an SSI than those without a reported allergy.
         iv. Adjust dose of antibiotics for obesity – See Table 2 for suggested dosing (ACS NSQIP measure).
         v. Patients with known nasal carriage of S. aureus should receive intranasal applications of mupirocin 2% ointment with or without a combination of chlorhexidine gluconate body wash (World Health Organization guidelines [WHO]).
         vi. Alcohol-based antiseptic solutions with chlorhexidine gluconate for surgical site skin preparation should be used in patients undergoing surgical procedures (WHO guidelines).
         vii. vii. Preoperative bowel preparation

(a) Although unaddressed by the Centers for Disease Control and Prevention (CDC), preoperative bowel preparation before colorectal surgery was recommended by the ACS/ Surgical Infection Society and WHO guidelines.
(b) A recent meta-analysis demonstrated that combined oral antibiotic prophylaxis combined with mechanical bowel preparation was associated with reduced SSI rates compared with mechanical bowel preparation alone. Oral antibiotic prophylaxis was associated with significant decreases in rates of anastomotic leak and 30-day mortality without increasing rates of *Clostridioides difficile* infections.
(c) Combined bowel preparation with oral antibiotic prophylaxis and mechanical bowel preparation is recommended by the 2019 American Society of Colon and Rectal Surgeons Clinical Practice Guidelines.

b. Intraoperative
   i. Re-dose antibiotics, if necessary – See Table 2 for dosing schedule during surgery (ACS NSQIP measure).
   ii. In general, antibiotics with short half-lives should be re-dosed at a frequency of 2 times the half-life of the agent. Antibiotics should also be re-dosed if there is significant intraoperative blood loss.
   iii. Goal of re-dosing is to maintain bactericidal concentrations throughout the operation. It may be prudent to consider re-dosing prophylaxis intraoperatively if large amounts of fluids and/or transfusions are being administered.
   iv. Maintain normothermia. Two small randomized controlled trials showed that maintaining normothermia reduced the risk of SSIs. The mechanism for this protective effect is currently unknown; however, a hypothesis is that this effect may result from impaired neutrophil function or subcutaneous vasoconstriction and subsequent tissue hypoxia with hypothermia.

c. Postoperative
   i. Discontinue prophylactic antibiotics within 24 hours after non-cardiac surgery and within 48 hours after cardiac surgery. Drains are not a sufficient reason to continue prophylactic antibiotics – ACS NSQIP measure.
   ii. Early recovery after surgery (ERAS) protocols: Aim to reduce the stress of surgery on patients by maintaining near normal physiology in the preoperative, intraoperative, and postoperative phases of care.
      (a) Early recovery after surgery protocols incorporate SSI prevention guideline recommendations, such as parenteral antibiotic prophylaxis and strict glycemic control, as well as interventions with newer evidence, such as oral antibiotic prophylaxis with mechanical bowel preparation, goal directed fluid therapy, and early enteral feeding.
      (b) A meta-analysis of 27 randomized controlled trials assessing 3279 patients undergoing abdominal/pelvic surgery showed a significant reduction in postoperative SSI for patients enrolled in ERAS pathways compared with conventional perioperative pathways.

5. Management of SSIs: See Infectious Diseases I chapter.

Table 1. Prophylactic Antibiotic Regimens for Surgery

Type of Surgery	Recommended Prophylaxis	Alternative Prophylaxis in Patients with β-Lactam Allergies
Coronary artery bypass grafting Other cardiac surgery Vascular surgery Hip arthroplasty Knee arthroplasty Neurosurgery	Cefazolin Cefuroxime	Vancomycin Clindamycin

Table 1. Prophylactic Antibiotic Regimens for Surgery *(continued)*

Type of Surgery	Recommended Prophylaxis	Alternative Prophylaxis in Patients with β-Lactam Allergies
Thoracic surgery	Cefazolin Ampicillin/sulbactam	Vancomycin Clindamycin
Gastroduodenal surgery	Cefazolin	Clindamycin *or* vancomycin + aminoglycoside *or* fluoroquinolone *or* aztreonam
Colorectal surgery	Cefotetan *or* cefoxitin or ampicillin/sulbactam *or* ertapenem Metronidazole + cefazolin *or* cefuroxime *or* ceftriaxone	Clindamycin + aminoglycoside *or* fluoroquinolone *or* aztreonam Metronidazole + aminoglycoside *or* fluoroquinolone
Abdominal hysterectomy Vaginal hysterectomy Biliary tract surgery	Cefotetan *or* cefoxitin *or* ampicillin/sulbactam	Clindamycin *or* vancomycin + aminoglycoside *or* fluoroquinolone or aztreonam Metronidazole + aminoglycoside *or* fluoroquinolone

Table 2. Recommended Dose and Dosing Interval for Commonly Used Antibiotics for Surgical Prophylaxis

Antimicrobial	Recommended Dose in Adults	Dosing Interval (hr)[a]
Ampicillin/sulbactam	3 g	2
Ampicillin	2 g	2
Aztreonam	2 g	4
Cefazolin	2 g, 3 g for patients weighing ≥ 120 kg	4
Cefuroxime	1.5 g	4
Cefoxitin	2 g	2
Cefotetan	2 g	6
Ceftriaxone	2 g	N/A[b]
Ciprofloxacin	400 mg	N/A[b]
Clindamycin	900 mg	6
Ertapenem	1 g	N/A[b]
Gentamicin	5 mg/kg of ideal or adjusted body weight[c]	N/A[b]
Levofloxacin	500 mg	N/A[b]
Metronidazole	500 mg	N/A[b]
Vancomycin	15 mg/kg	N/A[b]

[a]With presumed normal renal function and dose as needed

[b]One dose of the antibiotic should suffice for the duration of most surgical procedures.

[c]Use adjusted body weight if actual body weight is > 20% of ideal body weight (IBW). Adjusted body weight = IBW + 0.4 (actual weight − IBW).

N/A = not applicable.

E. Prevention of Catheter-Associated Urinary Tract Infections (CAUTIs) – ACS NSQIP Measures
   1. Definition – CDC – See Figure 1.

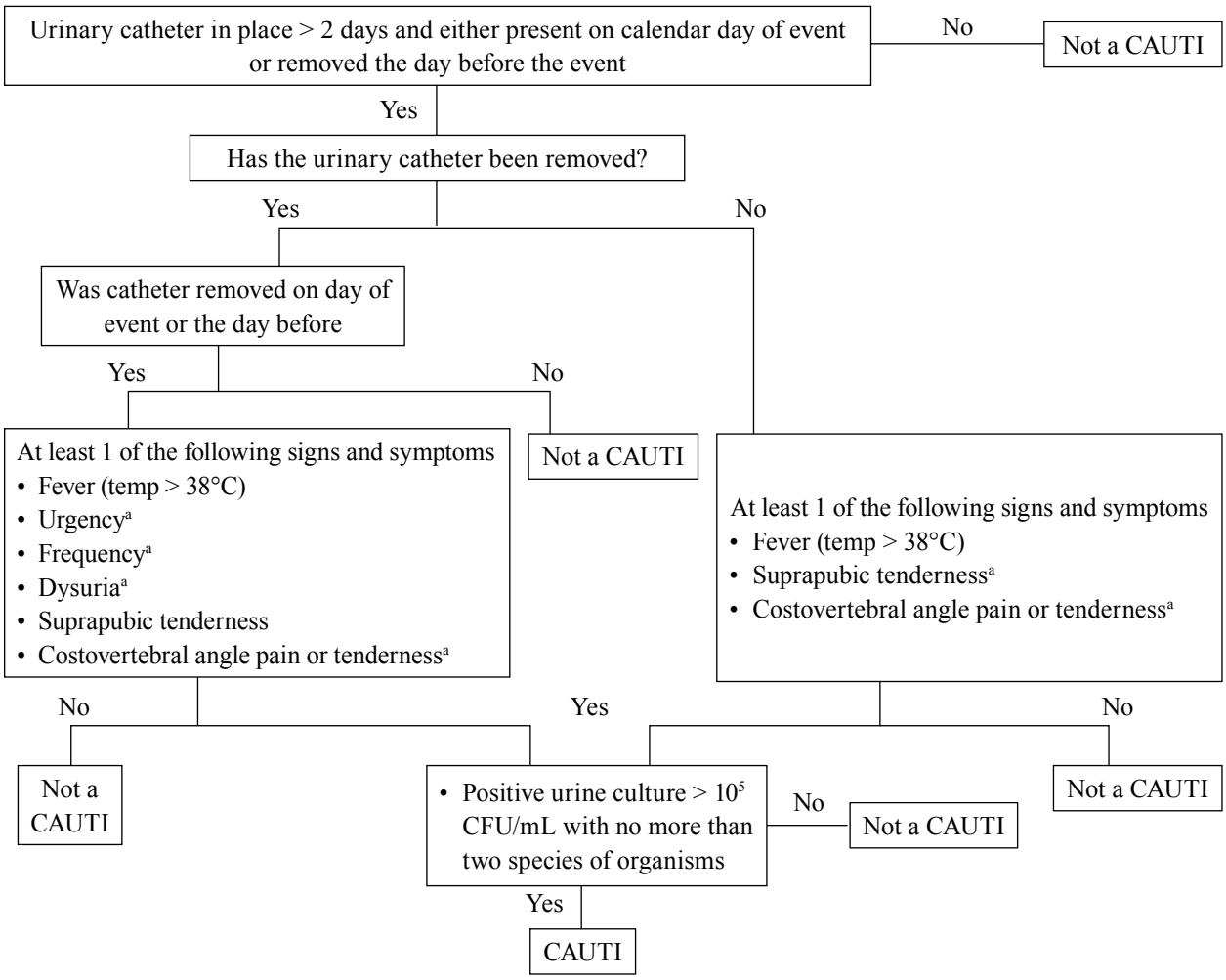


Figure 1. Diagnosis of catheter-associated urinary tract infection according to CDC criteria.

[a]With no other recognized cause

CAUTI = catheter-associated urinary tract infection.

2. Epidemiology and clinical significance. UTIs represent 30%–40% of all nosocomial infections. Up to 80% of UTIs are caused by urinary catheterization.
    a. More than 5% of postoperative patients will experience a UTI.
    b. CAUTIs account for $300–$400 million in additional health care costs per year.
    c. CMS no longer provides reimbursement to providers for the treatment of CAUTIs.
3. Risk factors: Azotemia, bacterial colonization of drainage bag, diabetes, female sex, increased duration of catheterization, and older age
4. Prevention strategies
    a. Minimize the use of prolonged urinary catheters. Target removal of urinary catheters as soon as possible postoperatively, preferably within 24 hours.
    b. Pharmacists may provide reminders as part of a multidisciplinary effort to minimize the placement and duration of urinary catheters.
    c. Strategies that should NOT be considered (all of these practices have low-quality evidence suggesting no benefits in preventing CAUTIs):

i. Routine use of silver or antibiotic-impregnated catheters. According to the CDC, silver or antibiotic-impregnated catheters can be considered if a comprehensive strategy to reduce CAUTI rates has failed. The ACS NSQIP best-practices guidelines suggest that use of antimicrobial urinary catheters can be considered for high-risk patients, such as those who may require prolonged (greater than 7–10 days) catheterization.
ii. Addition of antibiotics to drainage bag
iii. Use of systemic antibiotic prophylaxis solely due to catheter placement
iv. Routine screening or treatment of asymptomatic bacteriuria
v. Bladder irrigation with antibiotics

5. Management of CAUTIs: See Infectious Diseases I chapter.

Patient Case

1. C.L is a 55-year-old man admitted to the surgical ICU after exploratory laparotomy for duodenal perforation and peritonitis. The patient was transferred from the operating room to the surgical ICU with a urinary catheter in place. On day 4 of the surgical ICU stay, the patient is extubated, and his urinary catheter is removed. On day 6 of the surgical ICU stay, the patient develops signs and symptoms of UTI with fever, urgency, and frequency. Urinalysis reveals no nitrites or leukocyte esterase. Urine cultures show 100,000 CFU/mL of *E. coli*. You are asked by the quality officer of the surgical ICU to discuss this case and determine whether this patient will qualify for the definition of CAUTI according to CDC criteria. Which statement regarding the diagnosis of CAUTI in this patient is most appropriate?

A. Qualifies for definition of CAUTI because the patient is symptomatic and has positive cultures.
B. Does not qualify for definition of CAUTI because the patient had a negative urinalysis.
C. Qualifies for definition of CAUTI because the patient is symptomatic and had recent urinary catheter.
D. Does not qualify for definition of CAUTI because the catheter was removed 2 days before symptoms.

F. Prevention of Catheter-Related Bloodstream Infections – ACS NSQIP
1. Definition (CDC): Must satisfy both criteria
a. Laboratory-confirmed bloodstream infection
i. Definition 1: A recognized pathogen (not a common skin contaminant) found in one or more blood cultures AND the organism is not related to an infection at a separate site
ii. Definition 2: At least one sign or symptom of infection (temperature greater than 100.4°F [38°C], chills, hypotension) AND a common skin flora is cultured from two or more blood samples AND organism is not related to an infection at a separate site
b. Central line placed for more than 2 days before the date of event and central line must be present or removed within 24 hours of blood cultures being obtained.
2. Epidemiology and clinical significance
a. Incidence of catheter-related bloodstream infections: 2.7–7.4 infections per 1000 catheter-days
b. Accounts for 14,000–28,000 deaths per year
c. Increases mean length of stay (LOS) by 7 days
d. CMS no longer provides reimbursement to providers for the treatment of catheter-related bloodstream infections.
3. Risk factors
a. Patient factors: Diabetes, hypotension, male sex, neutropenia, parenteral nutrition, prolonged hospitalization before catheter placement

   b. Catheter-related factors: Heavy microbial colonization at insertion site, inadequate catheter care, internal jugular or femoral vein placement site (compared with subclavian vein), prolonged duration of venous catheterization, use of non-tunneled catheters
4. Prevention strategies
   a. Use appropriate techniques for catheter insertion (e.g., skin preparation and maximal sterile barrier precautions) and management (e.g., daily site care, routine site evaluation for local erythema). Minimize the use of central venous catheters. Pharmacists may provide reminders as part of a multidisciplinary effort to minimize the placement and duration of venous catheters.
   b. Consider antibiotic lock therapy in patients with limited venous access or a history of catheter-related bloodstream infections.
   c. Do not use systemic antibiotic prophylaxis.
   d. Antimicrobial/silver-impregnated catheter should be considered in patients with an anticipated catheter duration greater than 5 days if rates of catheter-related bloodstream infections have not decreased despite the implementation of a comprehensive prevention strategy.
5. Management: See Infectious Diseases I chapter.

II. BACTERIAL MENINGITIS

A. Definitions
   1. Bacterial meningitis is a neurologic emergency involving mild to severe inflammation of the meningeal layers encasing the central nervous system (CNS). Cerebrospinal fluid (CSF) is intimate to the meninges and serves as both a medium for pathogen growth and a diagnostic fluid.
   2. Community-acquired meningitis is an infection unrelated to a neurosurgical procedure, neurotrauma, or hospitalization.
   3. Health care–associated or nosocomial meningitis is an infection related to invasive procedures, including craniotomy, internal or external ventricular catheters, lumbar puncture, intrathecal medication administration, and/or spinal anesthesia. Additional causes include complicated cranial trauma, traumatic brain injury, and hematogenous spread in patients with hospital-acquired bacteremia.

B. Epidemiology
   1. Bacterial meningitis has community-acquired or health care–associated epidemiology; health care–associated meningitis is usually associated with neurotrauma or neurosurgical procedures.
   2. Community-acquired bacterial meningitis has an annual incidence in adults of about two cases per 100,000 people. The incidence has been decreasing in recent years (two cases per 100,000 in 1998 to 1.38 cases per 100,000 in 2006; 31% decrease), likely because of increased vaccination against common pathogens. The incidence of nosocomial bacterial meningitis varies depending on the mechanism of neuro-anatomic disruption and ranges from 1.5% of patients undergoing craniotomy to 25% of post-trauma patients with basilar skull fracture.
   3. Delayed CSF sterilization beyond 24 hours is a risk factor for subsequent neurologic sequelae, including intracranial hypertension, seizures, and permanent neurologic deficit. Clinical presentations of septic shock, altered mental status, and seizures are associated with worse outcomes. Additional complications include respiratory failure and hyponatremia.
   4. Crude mortality for community-acquired meningitis is 19%–37%, whereas mortality for health care–associated meningitis is generally lower, particularly if associated with a reversible procedure or removable device.

C. Diagnosis
   1. The clinical diagnosis of meningitis is nonspecific and difficult to distinguish from that of other infections. Although headache, fever, neck stiffness, and altered mental status are present in almost 95% of patients with community-acquired meningitis, fever and a decreased level of consciousness are the most consistent clinical features in patients with health care–associated meningitis.
   2. Lumbar puncture or other method (e.g., from existing drain or shunt) to sample CSF for cell count and analysis, as well as Gram stain and culture, is necessary for definitive diagnosis. Neuroimaging with head computed tomography to detect prelumbar brain shift and risk of brain herniation should be done before lumbar puncture in patients with suspected cranial mass (e.g., immunosuppressed, papilledema, history of CNS disease, new-onset seizure, and focal neurologic deficit).
      a. Opening pressure during lumbar puncture is usually increased in bacterial meningitis (normal opening pressure in adults: 10–20 cm H_2O).
      b. Cell count and fluid analysis
         i. Community-acquired bacterial meningitis can be differentiated from other causes of meningitis (e.g., viral, aseptic). In general, bacterial meningitis is associated with CSF that is predominantly neutrophilic and has lower glucose concentration. Strong predictors of bacterial meningitis include:
            (a) CSF glucose less than 34 mg/dL
            (b) Ratio of CSF to blood glucose less than 0.23
            (c) CSF protein greater than 220 mg/dL
            (d) CSF leukocyte count greater than 2000 cells/mm^3
            (e) CSF neutrophil count greater than 1180 cells/mm^3
         ii. The diagnostic utility of CSF cell count and fluid analysis in health care–associated meningitis is unknown but is likely limited because of concomitant reasons for local inflammation related to devices or recent procedures. In the presence of CSF drains or recent history of neurosurgery, new headache, nausea, lethargy, and/or change in mental status may be a sign of new CNS infection. Neither normal CSF analysis nor negative Gram stain may exclude the presence of an infection. Elevated CSF lactate concentration may be useful to distinguish meningitis from other infectious sources.
      c. Gram stain and culture: Bacteriologic examination of CSF can provide rapid and reliable identification of the causative pathogen(s). Although CSF cell count and analysis is the diagnostic foundation for community-acquired meningitis, CSF culture is the most specific test for health care–associated meningitis. Of note, immunocompromised patients with meningitis may present with normal CSF cell counts.

D. Management and Treatment
   1. Appropriate empiric antibiotic therapy administered intravenously, targeted against likely pathogens, and guided by local antibiotic susceptibility patterns should be initiated as soon as possible after bacterial meningitis is suspected. See Table 3 for common pathogens and recommended empiric therapy.
   2. Antibiotic dosing and administration strategies should be chosen to optimize CSF concentrations relative to bacterial minimum inhibitory concentration (MIC) and antibiotic pharmacodynamic (PD) properties. Although meningeal inflammation may promote CSF penetration, data are inconsistent regarding its impact on effective antimicrobial delivery. Furthermore, as meningitis improves, inflammation will likely improve, and there is currently no way to predict the level of penetration in correspondence to the severity of the inflammation.
   3. Gram stain can be used to broaden bacterial coverage, but final culture should be reserved for antibiotic de-escalation and definitive antibiotic regimen.

4. Role of corticosteroids
    a. Adjunctive corticosteroids may improve outcomes by reducing reactive meningeal inflammation and neurologic sequelae related to antibiotic-induced bacterial lysis.
    b. Conflicting results are published regarding the effects of systemic corticosteroids on neurologic sequelae and mortality among patients with bacterial meningitis.
    c. Studies from high-income countries tend to suggest that systemic corticosteroids decrease or trend toward a decrease in mortality and neurologic sequelae.
    d. The outcome benefit associated with systemic corticosteroids seems most pronounced in patients with *Streptococcus pneumoniae* meningitis. However, because corticosteroids must be administered before the receipt of antimicrobials, it is unlikely that clinicians will know the etiology of the disease when making the decision for steroids.
    e. The Infectious Diseases Society of America (IDSA) guidelines recommend administering dexamethasone 0.15 mg/kg every 6 hours for up to 96 hours, with the first dose administered 10–20 minutes before, or at least concomitant with, the first dose of antimicrobial therapy. The IDSA guidelines also recommend for continuation of dexamethasone only if cultures show the presence of *S. pneumoniae*, although this recommendation is not supported by strong clinical evidence.
5. Health care–associated meningitis
    a. See Table 3 for recommended empiric treatment of health care–associated meningitis.
    b. In patients with CSF shunts, complete removal of an infected CSF shunt and replacement with an external ventricular drain combined with intravenous antimicrobial therapy is recommended.
    c. In selected patients with bacterial meningitis after placing a CSF shunt, the IDSA recommends direct instillation of antimicrobial agents intraventricularly through either an external ventriculostomy or shunt reservoir. This practice should only be considered in patients with pathogens that are difficult to eradicate or for those who cannot undergo catheter replacement. See Table 4 for selected antimicrobial intraventricular dosing. When intraventricular antibiotics are administered, drains should be clamped for 15-60 minutes to allow antibiotics to equilibrate throughout the CSF. Antimicrobial agents administered into the CNS should be preservative free. Dosing and intervals of intraventricular antimicrobial therapy should be adjusted on the basis of the CSF antimicrobial concentration being 10-20 times the MIC of causative pathogen, ventricular size, and daily output from ventricular drain. Given the paucity of pharmacokinetic data surrounding intraventricular antibiotics administration and the lack of head-to-head comparisons with intravenous antimicrobials, the administration of systemic antimicrobials should be continued in conjunction with intraventricular antibiotics.
6. Definitive antibiotic therapy can be determined according to finalized cultures and susceptibility results. In general, definitive therapy should entail choosing the therapy with the most appropriate spectrum of activity and an adequate penetration into the CSF. Therapy should be continued for at least 7 days in all patients with meningitis. *S. pneumoniae*, coagulase-negative staphylococci, *S. aureus*, gram-negative bacilli, and *Cutibacterium acnes* (previously known as *Propionibacterium acnes*) should be treated for 10–14 days and *Listeria monocytogenes* for at least 21 days. CSF should be negative for *S. aureus* for at least 10 days before shunt replacement.

Table 3. Common Pathogens Seen in Different Bacterial Meningitis Populations and the Corresponding Recommended Empiric Therapy

Patient Group	Common Pathogens	Recommended Empiric Therapy
Community-Acquired Pathogens		
18–50 yr	*N. meningitidis*, *S. pneumoniae*	Vancomycin + third-generation cephalosporin (e.g., ceftriaxone)
> 50 yr *or* any age with predisposing condition[a]	*N. meningitidis*, *S. pneumoniae*, *L. monocytogenes, H. influenzae,* aerobic GNB (e.g., *E. coli)*	Vancomycin + third-generation cephalosporin (e.g., ceftriaxone) + ampicillin
Health care–associated or nosocomial		
Basilar skull fracture (communication with sinuses or oropharynx)	*S. pneumoniae*; *H. influenzae*; group A β-hemolytic streptococci	Vancomycin + third-generation cephalosporin (e.g., ceftriaxone)
Health care–associated or nosocomial		
Penetrating trauma	*S. aureus*, CoNS, aerobic enteric (e.g., *E. coli*) and non–lactose-fermenting (e.g., *P. aeruginosa*) gram-negative bacilli	Vancomycin + cefepime *or* ceftazidime *or* meropenem
Post-neurosurgery	Aerobic enteric (e.g., *E. coli*) and non–lactose-fermenting (e.g., *P. aeruginosa*) gram-negative bacilli, *S. aureus*, CoNS	Vancomycin + cefepime *or* ceftazidime *or* meropenem
CSF shunt	Aerobic enteric (e.g., *E. coli*) and non–lactose-fermenting (e.g., *P. aeruginosa*) gram-negative bacilli, *S. aureus*, *Propionibacterium acnes*	Vancomycin + cefepime *or* ceftazidime *or* meropenem

[a]Altered immune status, alcoholism

CoNS = coagulase-negative staphylococci; GNB = gram-negative bacteria.

Table 4. Selected Antibiotic CSF Penetration and Dosages for Meningitis

Agent	CSF Penetration, % of Serum Concentration	Dosage for Meningitis
Amikacin	Low	Conventional: 10–15 mg/kg[a] IV q8–12hr Extended interval: 20–25 mg/kg[a] IV once daily Intraventricular: 30 mg once daily
Ampicillin	13–14	2 g IV q4hr
Cefazolin	8–15	2 g IV q8hr
Cefepime	10–20	2 g IV q8hr
Cefotaxime	10–12	2 g IV q4hr
Ceftazidime	30–40	2 g IV q8hr
Ceftriaxone	8–12	2 g IV q12hr
Ciprofloxacin	20–25	400 mg IV q8hr
Colistin	Low	Intraventricular: 10 mg once daily
Meropenem	6–21	2 g IV q8hr (extended infusions over 3 hr may be considered)

Table 4. Selected Antibiotic CSF Penetration and Dosages for Meningitis *(continued)*

Agent	CSF Penetration, % of Serum Concentration	Dosage for Meningitis
Nafcillin	1–2.5	2 g IV q4hr
Piperacillin/tazobactam	20–30	Not recommended because of low penetration of tazobactam
Tobramycin	10–20	Conventional 2–3 mg/kg IV q8–12hr Extended interval: 5–7 mg/kg IV once daily Intraventricular 10–20 mg once daily
Vancomycin	30	15–20 mg/kg IV q8–12hr Intraventricular 10–20 mg once daily

[a] Use IBW unless actual body weight is > 20% of IBW. If actual body weight is >20% IBW use adjusted body weight. Adjusted body weight = IBW + 0.4 (actual weight − IBW).

IV = intravenous(ly); q = every

Patient Case

2. H.K. is a 42-year-old woman who presents with her husband to the emergency department from home with mental status changes, lethargy, and a temperature of 102.9°F (39.4°C). Her husband reports that she had a severe headache 24 hours earlier. She has no significant medical history. Diagnostic workup results are highly suggestive of meningitis. A lumbar puncture is performed, during which a high initial pressure is noted. Which intervention would be best initially for H.K.?
   A. Administer dexamethasone 0.15 mg/kg intravenously.
   B. Await laboratory and microbiologic analysis.
   C. Initiate empiric ceftriaxone and vancomycin.
   D. Initiate empiric ampicillin, cefepime, and vancomycin.

III. ANTIMICROBIAL STEWARDSHIP

A. Goals: To optimize clinical outcomes while minimizing unintended consequences of antimicrobial use, which include the emergence of antimicrobial resistance and adverse drug reactions. In addition, a responsible approach to the use of antimicrobial agents should reduce the overall costs associated with treatment.
   1. Antibiotics are the second most common class of drugs that cause adverse effects.
   2. Antibiotics are the most common class of medication to be associated with prescribing errors.
   3. Minimizing antimicrobial resistance by optimizing therapy should lead to measurable benefits at the patient level because studies have shown the negative effect of antimicrobial resistance on several clinical outcomes.
   4. In 2018, CMS finalized a rule that requires all acute care hospitals participating in Medicaid/Medicare services to implement antimicrobial stewardship programs.

5. In 2019, the CDC updated the hospital core elements to reflect the new evidence in the field of antimicrobial stewardship. Major updates include hospital leadership commitment, accountability, pharmacy expertise, tracking, reporting, and education.

B. Antimicrobial Stewardship Definition: Multidisciplinary and coordinated interventions designed to improve and measure the appropriate use of antimicrobial agents and to promote the selection of optimal antimicrobial drug regimen, including dosing, duration, and route of administration.
   1. Core members of a multidisciplinary antimicrobial stewardship team should include an infectious diseases physician, a clinical pharmacist, a clinical microbiologist, an information systems specialist, an infection control professional, a nurse, and a hospital epidemiologist.
   2. Broad-spectrum antimicrobials are often necessary in the ICU to provide early effective therapy. However, continued use of broad-spectrum antimicrobials may lead to the selection of pathogenic organisms (i.e., *C. difficile, Candida* spp.), nephrotoxicity, and the emergence of resistance.

C. Critical Care Pharmacists – Well positioned to provide guidance on antimicrobial therapy, including expedited selection of appropriate initial agents, aggressive dosing to optimize PD, interpretation of microbiological evidence, appropriate de-escalation of antimicrobials, monitoring of response and potential adverse effects, and determination of the appropriate treatment duration
   1. Critical care pharmacists could either have an active role as a member of a multidisciplinary antibiotic stewardship team or serve in many of the different roles and activities of antibiotic stewardship.
   2. Core activities that can be pursued through formal pathways or everyday clinical interventions:
      a. Education with active intervention – Interpretation of rapid identification tests and susceptibility testing
      b. Guidelines and clinical pathway development. Recent IDSA guidelines recommend facility-specific clinical practice guidelines or algorithms as an effective way of standardizing prescribing practices.
      c. Streamlining or de-escalation of therapy
         i. Shortening therapy duration
         ii. Discontinuing unnecessary antimicrobials
      d. Dose optimization
         i. Application of pharmacokinetic (PK)/PD principles. Per IDSA recommendations, PK and dose-monitoring programs are specifically recommended for vancomycin and aminoglycosides.
         ii. Dose adjustments based on organ function
         iii. Allergy detection and assessment, in particular promotion of β-lactam skin testing when appropriate
      e. Parenteral to oral conversion
      f. Recent IDSA guidelines recommend against the use of didactic education alone for stewardship.
   3. Support from microbiology laboratory and electronic medical record system surveillance and clinical decision support may further enhance stewardship efforts.

D. Antimicrobial Stewardship Strategies (can be used synergistically)
   1. Prospective audit with intervention and feedback (“back-end” strategy)
      a. Allows flexibility and minimizes delay in administering therapy
      b. The most successful strategy involves direct communication with treating physicians and required documentation for acceptance of recommendation or rationale for denial.
   2. Formulary restriction and preauthorization (“front-end” strategy).
      a. May be resource-intensive, and prescribers may feel a loss of autonomy
      b. Initial choices of antimicrobials may be optimized through consultation with infectious disease experts.

   c. Antimicrobial cycling: An example of formulary restriction in which there is a scheduled removal and substitution of a specific antimicrobial or an antimicrobial class. It is an effort to minimize antimicrobial selection pressures. Evidence is insufficient to suggest that antimicrobial cycling strategies are effective. IDSA stewardship guidelines recommend against the use of antimicrobial cycling as a stewardship strategy. The compensatory overuse of another class of antibiotics has been reported when attempts to restrict a different class of antibiotics were implemented.
3. Regardless of the strategies used or the quality of clinical pathways, programmatic antibiotic stewardship is not a substitute for clinical judgment.
4. Process indicators (e.g., days of therapy, provider adherence to clinical pathway, time to effective therapy) and outcome measures (e.g., length of stay, mortality, 30-day readmission) should be used to gauge program success.

E. Effective Antimicrobial Stewardship Programs Can Be Financially Self-Supporting
1. Antimicrobials can account for up to 30% of hospital pharmacy budgets, with up to 50% of antimicrobial use being inappropriate, leading to increased cost, increased selection of resistant pathogens, and increased selection of opportunistic infections (OIs).
2. Several studies have shown significant cost savings associated with antimicrobial stewardship programs. However, most of these studies focused only on antimicrobial costs, with little focus devoted to indirect costs and implementation costs.
3. There is currently a proposed CMS rule for incorporating antibiotic prescribing patterns as one of the quality-reporting metrics. Many CMS quality metrics are tied to reimbursement. If this metric becomes tied to reimbursement, there will be further incentives to implement antimicrobial stewardship programs.
4. Effective January 2017, the Joint Commission announced a new medication management standard that addresses antimicrobial stewardship. Many of the standards are applicable and relevant to critical care pharmacists: continuing education of staff, education of patients and family members, participation in an antimicrobial stewardship team, and development and execution of antimicrobial management protocols.

F. Evidence Supporting Antimicrobial Stewardship Efforts in Critically Ill Patients
1. Studies in this area have been limited by poor study design. However, most studies have reported a decreased use in either antibiotics overall or a targeted class of antibiotics. Some studies have also reported a decrease in key resistance rates.
2. Antimicrobial stewardship strategies used in the studies have varied widely, including restriction, formal infectious disease physician consultation, protocols for de-escalation, implementation of computer-assisted decision support, and formal reassessments of the empiric antibiotics by a stewardship team.
3. Meta-analyses of before-after studies evaluating prospective audit and feedback to ICU patients receiving antibiotics found no increase in mortality, suggesting that this practice could safely be implemented in critically ill patients.

IV. RAPID DIAGNOSTIC TESTS

A. Rationale
   1. There is an association between delay in administration of appropriate antimicrobials and decreased survival among patients with septic shock.
   2. Broad-spectrum antimicrobials are usually used in severe infections to ensure that all potential pathogens are covered.
   3. Rapid diagnostic tests may assist in de-escalation efforts in an attempt to practice antimicrobial stewardship. Critical care pharmacists are advocates for the appropriate use of antimicrobials according to the results of rapid diagnostic tests.
   4. In many cases, the implementation of rapid diagnostic tests may be cost neutral, or even constitute a cost savings, when antimicrobial stewardship efforts leading to decreased consumption of antibiotics occur.
   5. Recommended by IDSA antimicrobial stewardship guidelines to be used in combination with stewardship team to optimize antibiotic therapy and improve clinical outcomes
   6. The Society of Infectious Diseases Pharmacists released an official position statement stating that rapid diagnostic tests can help antimicrobial stewardship programs decrease unnecessary exposure and optimize patient care.

B. Early Pathogen Identification from Positive Bloodstream Cultures
   1. All discussed methods attempt to shorten the time from blood culture positivity to species identification or susceptibility testing. Traditional pathogen identification and susceptibility testing can take 72–96 hours. Early pathogen identification techniques seek to provide clinically actionable information within the first 24 hours from the time of culture positivity. See Table 5 for examples of rapid diagnostic tests, corresponding targets, and respective turnaround times.
   2. Most of these techniques, when combined with antibiotic stewardship efforts, lead to significant reductions in health care costs and improvements in clinical outcomes in relevant studies. However, many of these were single-center studies, which makes external validity questionable. The overall impact of the implementation of rapid diagnostic tests in a single institution is determined by several factors.
      a. Epidemiology of targeted organisms
      b. Presence and actions of existing antimicrobial stewardship teams
      c. Current clinician prescribing patterns
      d. Patient population
   3. The inability to detect polymicrobial infections is a common limitation to most of the techniques described.
   4. Newer methods using whole blood (vs. blood culture medium or agar plates) have been developed (LightCycler SeptiFast, SepsiTest).
      a. A benefit to identification from whole blood is the ability to identify bacteria in patients with recent or current antibiotic exposure.
   5. Some of the technology discussed has been developed for respiratory cultures, but the clinical adaptation in that arena is considerably less than that for blood cultures.
   6. Application of rapid diagnostic tests that detect genetic encoding of resistance mechanisms requires further education and guidelines to assist clinicians in choosing the proper therapy because traditional susceptibility results are not available. See Table 6 for a reasonable approach based on the detection of resistance genes and species.
   7. Peptide nucleic acid (PNA) fluorescence in situ hybridization (FISH)
      a. Mechanism: Targets species-specific ribosomal RNA from positive blood cultures
      b. Sensitivity and specificity: 96%–100%

c. Limitations: Does not provide antimicrobial sensitivity data. Currently available FISH products are only used for species identification; however, the FDA approved a new product that could detect the *mecA* gene for detecting the presence of methicillin resistance. This product is not currently commercially available.
d. Application
    i. Separates *S. aureus* from possible skin flora contamination of coagulase-negative staphylococci (CoNS).
    ii. Differentiates *Enterococcus faecium* (which is often resistant to ampicillin and vancomycin) from *Enterococcus faecalis*.
    iii. Identifies fluconazole-sensitive *Candida* spp. for patients empirically treated with echinocandins.
    iv. Detects *Pseudomonas* versus non-*Pseudomonas* gram-negative spp. in patients treated with combination gram-negative therapy.
e. Studies
    i. A retrospective study evaluating the outcome and economic benefit of PNA FISH methods for the early differentiation of CoNS and *S. aureus* bacteremia in clinical practice showed a significant cost savings and a decrease in median LOS. Of note, the PNA FISH results were combined with the efforts of an antimicrobial therapy team. Similar results have been shown with PNA FISH implementations for other pathogens, including *Enterococcus* spp. and *Candida* spp. These results are in contrast to those of another study that evaluated the pre- and post-staphylococci PNA FISH implementation results without the use of an antimicrobial stewardship team. This study found no significant effects on patient LOS or vancomycin use. This suggests that rapid identification tests are probably beneficial only when combined with educational efforts and prospective alerts to notify clinicians of the clinical applicability of the test results.
    ii. In a prospective randomized controlled study in which patients were randomized to early notification of PNA FISH results for CoNS or *S. aureus* within 3 hours or usual care, intervention was associated with decreased mortality, decreased antibiotic use, and decreased LOS. The most pronounced benefits occurred in critically ill patients.

8. Mass spectroscopy: Matrix-assisted laser desorption-ionization/time of flight (MALDI-TOF)
    a. Mechanism: Mass spectroscopy is compared with library standards for identifying pathogen species and/or resistance mechanisms. Mass spectroscopy is an analytic technique whereby samples are ionized that produces a mass spectrogram for the sample. Each sample has a unique mass spectrogram, similar to a fingerprint, which can be matched to a library of reference standards for identification.
    b. Sensitivity and specificity: 98%–100%
    c. Limitations
        i. Similar to PNA FISH, no antimicrobial susceptibility is reported; however, the technology could detect genes encoding resistance. MALDI-TOF for resistance genes is currently not commercially available.
        ii. Cannot be used for polymicrobial cultures
        iii. No library is available for unusual organisms, though pathogen libraries are consistently being updated.
    d. Application: Potentially wider clinical applicability than PNA FISH (which is limited by the availability of specific tests for certain pathogens) with early identification of many more pathogens. Additional library standards for different pathogens are continually being added.

e. Studies: This strategy has been evaluated in several studies. In a before-and-after study of MALDI-TOF implementation, a shortened time to pathogen identification and a decrease in LOS, recurrent infections, and mortality were seen. Of note, the implementation of MALDI-TOF included communicating the results to the treating clinicians by an antimicrobial stewardship team with evidence-based antibiotic recommendations.

9. Polymerase chain reaction (PCR)-based detection systems
   a. MRSA PCR test
      i. Mechanism: Novel multiplex real-time assay for *mecA* gene
      ii. Sensitivity and specificity: 85% and 92%, respectively, for MRSA and methicillin-susceptible *S. aureus* (MSSA) identification
      iii. Application: Earlier de-escalation of anti-MRSA antimicrobials and earlier appropriate treatment with antistaphylococcal penicillin (oxacillin, nafcillin, cefazolin) for MSSA
         (a) Most experience and evidence with using the MRSA PCR test is for de-escalation of MRSA-active antibiotics in patients with pneumonia. New evidence also shows this may be used for de-escalation in patients with intra-abdominal infections, bloodstream infections, wound cultures, and urinary cultures.
         (b) Providers may use results from MRSA PCR test for up to 14 days between specimen collections (negative predictive value at 14 days in one study was 95.5%). Another study demonstrated that MRSA nares screening within 60 days of ICU admission has a high negative predictive value (98%) and specificity (96%).
      iv. Studies: In a before-and-after study, implementation of the MRSA PCR test resulted in reduced time to appropriate therapy and duration of unnecessary MRSA coverage. In addition, the mean hospital costs were decreased, and there was a trend toward decreased LOS. In a very similar study, the combination of the MRSA PCR test with antimicrobial stewardship efforts resulted in significant decreases in LOS and cost and a trend toward decreased mortality (18% vs. 26%).
   b. FilmArray System
      i. Mechanism: Uses multiplex PCR technology to identify bacterial species and resistance genes
      ii. Sensitivity and specificity
         (a) For species identification: Greater than 90%
         (b) For resistance genes: 100% (currently only available for *mecA* - methicillin resistance; van A/B - vancomycin resistance; *Klebsiella pneumoniae* carbapenemase [KPC] - carbapenem resistance)
      iii. Application: Earlier escalation or de-escalation of antimicrobial agents
      iv. Studies have shown an earlier time to pathogen identification and antibiotic de-escalation. One study found no difference in mortality, LOS, or cost when multiplex PCR was combined with antimicrobial stewardship.

10. Nanoparticles (Microarray): Verigene blood culture test
    a. Mechanism
       i. Direct detection from positive blood culture medium using nanoparticle technology
       ii. Nucleic acid extraction and array hybridization
    b. Most-developed commercially available product with resistance gene detection
    c. Sensitivity and specificity: 93%–100%
    d. Studies: Currently, most studies are limited to in vitro evaluations, which showed high levels of accuracy and decreased time to pathogen and resistance mechanism identification. Limited clinical evidence; however, two small single-center studies have evaluated the use of Nanosphere technology to augment clinical decisions. These studies showed a decrease in time to appropriate antimicrobials, in addition to decreased LOS and overall cost.

11. Chromogenic media
    a. Mechanism
        i. Microbiological media used to identify different microorganisms by color production
        ii. Growth media use enzyme substrates that release colored dyes on hydrolysis, with a wide range of enzymes that can be targeted
        iii. Potential advantage of being able to detect polymicrobial growth
    b. Sensitivity and specificity: 95%–100%
    c. Limitations
        i. Many different companies make different chromogenic agar media (Brilliance, chromID, CHROMagar). Slight differences in sensitivity and specificity were seen in studies; however, all were within acceptable ranges.
        ii. Time to identification is longer than with other rapid diagnostic tests.
        iii. Different manufacturers' chromogenic agar produces different colors for positive identification. Readers of chromogenic agar should be sufficiently trained and familiar with the product used by the local institution.
    d. Application
        i. Isolation of *S. aureus* from other *Staphylococcus* spp.
        ii. Detection of methicillin resistance among *S. aureus*
        iii. Detection of vancomycin resistance
        iv. Detection of specific Enterobacterales: *Salmonella*, *E. coli* O157, extended-spectrum β-lactamase (ESBL) production
        v. Differentiation of different *Candida* spp.
        vi. Detection of KPC
    e. Studies: Many clinical studies have shown significant advantages over conventional culture media. With the advent of newer technology and shorter detection times, the clinical applicability of chromogenic media may be limited. However, few microbiology laboratories have implemented rapid diagnostic methods because of the considerable upfront costs. Centers where chromogenic media are being used may continue to rely on this technology.

Table 5. Examples of Rapid Diagnostic Tests of Positive Blood Cultures and Their Characteristics

Assay Technology	Manufacturer/ Trade Name	Organisms/Antimicrobial Resistance Targets	Detection Time After Positive Growth on Blood Culture
PNA FISH	• AdvanDx/PNA FISH, Traffic Light • AdvanDx/ QuickFISH	*S. aureus*, CoNS, *E. faecalis*, *E. faecium*, *E. coli*, *K. pneumoniae*, *P. aeruginosa*, *C. albicans*, *C. glabrata*, *C. parapsilosis*, *C. tropicalis*, *C. krusei*	1.5 hr – PNA FISH 20 min – QuickFISH – Not currently FDA approved for *Candida* spp.
PCR	• BD GeneOhm/ Staph SR • Cepheid/Xpert MRSA	*S. aureus*, CoNS/*mecA* (methicillin)	1–2 hr

Table 5. Examples of Rapid Diagnostic Tests of Positive Blood Cultures and Their Characteristics[a] *(continued)*

Assay Technology	Manufacturer/ Trade Name	Organisms/Antimicrobial Resistance Targets	Detection Time After Positive Growth on Blood Culture
PCR	• BioFire/ FilmArray	*Enterococcus* spp., *L. monocytogenes* *S. aureus*, *S. agalactiae*, *S. pneumoniae*, *S. pyogenes*/*mecA* (methicillin), vanA/B (vancomycin) *A. baumannii*, *H. influenzae*, *N. meningitidis*, *P. aeruginosa*, *E. cloacae*, *E. coli*, *K. oxytoca*, *K. pneumoniae*, *Proteus* spp., *S. marcescens*/KPC (carbapenem) *C. albicans*, *C. glabrata*, *C. krusei*, *C. parapsilosis*, *C. tropicalis*	1 hr – Direct from blood culture media
MALDI-TOF	• BioMérieux/ Vitek MS • Bruker/ Microflex	Several pathogens, including bacteria, yeast, mold, mycobacteria/resistance mechanisms are in development	10–30 min
Nanoparticles (Microarray)	• Nanosphere/ Verigene	*S. aureus*, *S. epidermidis*, *S. lugdunensis*, *S. anginosus*, *S. agalactiae*, *S. pneumoniae*, *S. pyogenes*, *E. faecalis*, *E. faecium*, *Staphylococcus* spp., *Streptococcus* spp., *Micrococcus* spp., *Listeria* spp./*mecA* (methicillin), vanA (vancomycin), vanB (vancomycin) *E. coli*, *K. pneumoniae*, *K. oxytoca*, *P. aeruginosa*, *S. marcescens*, *Acinetobacter* spp., *Citrobacter* spp., *Enterobacter* spp., *Proteus* spp./ CTX-M (ESBL), IMP (carbapenemase), KPC (carbapenemase), NDM (carbapenemase), OXA (carbapenemase), VIM (carbapenemase) *Candida* spp. in development	2 hr – Direct from blood culture media
Chromogenic media	• CHROMagar	*S. aureus*, MRSA, *Enterococcus* spp.,VRE, *S. agalactiae, E. coli* (Shiga toxin, EC-O157), *Klebsiella* spp., *Proteus* spp., *Pseudomonas* spp., *Acinetobacter* spp., *Yersinia* spp.; KPC (carbapenem), CTX-M (ESBL) *C. albicans*, *C. tropicalis*, *C. krusei*	24–48 hr

ESBL = extended-spectrum β-lactamase; KPC = *Klebsiella pneumoniae* carbapenemase; MALDI-TOF = matrix-assisted laser desorption-ionization/time of flight; MRSA = methicillin-resistant *S. aureus*; NDM = New Delhi metallo-β-lactamase; PCR = polymerase chain reaction; PNA FISH = peptide nucleic acid fluorescence in situ hybridization; VRE = vancomycin-resistant enterococci.

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Table 6. Reasonable Empiric Treatment Approach Associated with Detection of Antimicrobial Resistance Encoding Genes from Rapid Diagnostic Tests for Positive Blood Cultures[a] *(continued)*

Resistance Genes	Species	Recommended Therapy	Alternative Therapy
mecA-negative	*S. aureus*	Oxacillin Nafcillin Cefazolin	Vancomycin (if intolerance to β-lactams)
mecA-positive	*S. aureus*	Vancomycin	Daptomycin Linezolid Ceftaroline
vanA/B-negative	*E. faecalis*	Ampicillin	Vancomycin (if allergic to β-lactams)
vanA/B-positive	*E. faecalis*	Daptomycin	Linezolid
vanA/B-negative	*E. faecium*	Vancomycin	Daptomycin (if intolerance to vancomycin)
vanA/B-positive	*E. faecium*	Daptomycin	Linezolid
CTX-M (ESBL)[a]	Enteric gram-negative pathogens	Carbapenems (ertapenem, imipenem, meropenem)	Ceftazidime/avibactam
KPC[a]	*K. pneumoniae* and other enteric gram-negative pathogens	Ceftazidime/avibactam Meropenem/vaborbactam Imipenem/cilastatin/relebactam	Colistin/polymyxin B Tigecycline/eravacycline
NDM carbapenemase[a]	*K. pneumoniae* and other enteric gram-negative pathogens	Cefiderocol Aztreonam + ceftazidime/avibactam	Colistin/polymyxin B Tigecycline
VIM or IMP carbapenemase[a]	*P. aeruginosa*	Colistin/polymyxin B Aztreonam Cefiderocol	
OXA β-lactamase[a]	*P. aeruginosa* and Enterobacterales	Colistin/polymyxin B Carbapenem[b]	Cefiderocol
OXA β-lactamase[a]	*A. baumannii*	Minocycline Ceftazidime/avibactam Sulbactam/durlobactam	Cefiderocol Colistin/polymyxin B Tigecycline Eravacycline Carbapenem

[a]Resistance mechanisms associated with gram-negative pathogens are more complicated. Often, there is more than one resistance mechanism. Hence, the recommended therapies represent reasonable empiric approaches before full antimicrobial susceptibility reports. However, therapy may be tailored to either a broader or a narrower spectrum.

[b]Carbapenem may have a higher MIC in the presence of OXA β-lactamase. If using carbapenem, may consider using maximal doses.

Patient Case

3. A 54-year-old man is admitted to the medical ICU with acute necrotizing pancreatitis. He is found to have an infected pancreatic abscess, which is being treated with meropenem and vancomycin. One week into the course, the patient develops a fever and leukocytosis. He also becomes hemodynamically unstable and oliguric. He is initiated on caspofungin empirically to cover for possible invasive candidiasis. Blood cultures are obtained, which become positive with a preliminary result of yeast. This institution is equipped with PNA FISH technology, which identifies the yeast as *Candida parapsilosis* on day 2 of therapy. The patient remains hemodynamically unstable. Which is the most appropriate choice for this patient's antifungal therapy?
   A. Continue caspofungin while fungal susceptibilities are finalized.
   B. Change to voriconazole 6 mg/kg every 12 hours.
   C. Change to liposomal amphotericin 5 mg/kg/day.
   D. Change to fluconazole 400 mg intravenously daily.

C. Early *C. difficile* Identification: Nucleic acid amplification methods (PCR and loop-mediated isothermal amplification)
   1. Assays are targeting DNA sequence for the toxin A or toxin B gene.
   2. Turnaround time: 1–3 hours
   3. Sensitivity and specificity: 90%–96%
      a. Because of increased sensitivity, the false-positive rates may be increased.
      b. As the prevalence of *C. difficile* decreases, the positive predictive value decreases, which may lead to unnecessary overtreatment.
      c. Educational efforts should be made to discourage the practice of over-ordering *C. difficile* rapid nucleic acid amplification tests. In addition, clinicians should be discouraged from ordering serial tests, which was a common practice when enzyme immunoassays were used. Because the sensitivity is sufficiently high, serial ordering only furthers the chance of false positivity.

D. Early Identification of Fungal Organisms
   1. Up to 50% of patients with histopathologically proven invasive candidiasis have negative blood cultures.
   2. Cultures for *Aspergillus* lack sensitivity and may require significant growth time.
   3. Biopsies are often not feasible in critically ill patients.
   4. Surrogate markers of fungal infections are needed for early diagnosis and to avoid delays in treatment, which have been associated with worsened outcomes.
   5. β-d-Glucan
      a. Found in the cell membrane of most fungal pathogens (except for *Mucor* and *Cryptococcus*). Potentially a screening tool for *Candida*, *Aspergillus*, and *Pneumocystis*.
      b. Fungitell assay can detect β-d-glucan in serum and provide results within 2 hours.
      c. Variable cutoffs are described in the literature for a positive result. A suggested cutoff for positive results, as used in a recent study evaluating prophylactic and preemptive antifungal therapy in critically ill patients, is greater than 80 pg/mL.
      d. Considerable differences in sensitivity and specificity values have been seen in different populations.
      e. Consistently high negative predictive values and low positive predictive values have been seen for the diagnosis of invasive fungal infections, which suggests that it is best used as a screening-out tool when the values are low. Low positive predictive value may occur because many clinical scenarios could lead to false-positive results.
         i. Exposure to β-d-glucan–containing surgical supplies and topical products

ii. Colonization with *Candida*
iii. Thrush and mucositis
iv. Cellulose membranes from dialysis filters
v. Bacterial infections
vi. Receipt of β-lactam antibiotics, albumin, or immunoglobulins

f. When used as a tool to potentially initiate treatment, optimal results occur when two consecutive tests are positive.
g. Nonspecific fungal element makes interpretation difficult. May detect elements from *Aspergillus*, *Candida*, and *Pneumocystis*. Does not detect different species.
h. Has been evaluated in combination with clinical prediction scores for invasive candidiasis in critically ill patients for initiating empiric antifungal therapy. When using β-d-glucan as a screening tool, empiric echinocandin therapy does not result in any significant differences in clinical outcomes compared with placebo.

6. T2 magnetic resonance
   a. Nanodiagnostic approach, which detects amplified *Candida* DNA. Similar to traditional MRI techniques, but done on a micro scale.
   b. Performed on whole blood, where magnetic particles that are coated with agents specific for binding to *Candida* DNA.
   c. When *Candida* is present in whole blood, it will bind to the particles and cluster, causing microscopic disruptions in the magnetic fields.
   d. Currently, FDA approved for 5 pathogenic *Candida* species *(C. albicans, C. glabrata, C. parapsillosis, C. tropicalis, C. kruseii)*
   e. High sensitivity (91%) and specificity (98%) but lacks clinical study application
7. Mannan/Anti-mannan
   a. Mannan is a polysaccharide component of the fungal cell wall that is specific to *Candida* spp.
   b. A commercially available latex agglutination and enzyme immunoassay exists for both mannan antigen and anti-mannan antibibodies that develop in response to mannan.
   c. Both tests are more specific than the β-d-glucan test, but they are not as sensitive.
   d. A meta-analysis that included studies in non-neutropenic critically ill patients found improved sensitivity and specificity of both tests when used in combination. It also demonstrated a varied sensitivity depending on the *Candida* spp (C. albicans with the highest and *C. parapsilosis* with the lowest).
8. Galactomannan
   a. Cell wall component of *Aspergillus* spp.
   b. Platelia enzyme immunoassay can detect galactomannan in serum or other sterile fluids (bronchoalveolar lavage) within 4 hours.
   c. Variable sensitivity and specificity, with generally better positive and negative predictive values for the detection of *Aspergillus* in the patient population with hematologic malignancies than in the solid organ transplant (SOT) population
   d. False positives can occur with the simultaneous administration of certain β-lactam antibiotics (piperacillin/tazobactam, amoxicillin/clavulanic acid) or the presence of other invasive mycoses (*Penicillium*, histoplasmosis, blastomycosis).
9. PCR
   a. An FDA-approved assay is available for detecting *Candida* spp. only.
   b. This test allows for the early detection of candidemia.
   c. This assay has high sensitivity and specificity and a good negative and positive predictive value.
   d. Colonization may decrease specificity and using the test too early (e.g., prior to onset of new symptoms) may decrease sensitivity.

E. Procalcitonin (PCT)
   1. An inflammatory biomarker that reflects host response to bacterial infections
   2. PCT synthesis is up-regulated by bacterial toxins and certain bacterial proinflammatory mediators such as interleukin (IL)-1b, IL-6, and tumor necrosis factor alpha (TNFα), but it is neutral to cytokines that are normally released for viral infections such as interferon-γ. Usual concentrations of PCT are undetectable (less than 0.05 mcg/L). However, on exposure to bacterial toxins, PCT is rapidly released within 2–4 hours. The plasma half-life of PCT is 24 hours. Concentrations in the literature for infected patients vary greatly; however, it appears that a higher max concentration of PCT during infection correlates with a higher incidence of mortality.
   3. PCT is used in many roles, including the diagnosis and prognostication for sepsis. IDSA stewardship guidelines recommend serial measurements be used in conjunction with other stewardship interventions to decrease antibiotics use. The Surviving Sepsis Campaign provides a weak recommendation (weak recommendation, low quality of evidence) for the use of low PCT to assist clinicians in the discontinuation of empiric antibiotics when no evidence of infection is found among patients with sepsis.
   4. For clinical decisions regarding antibiotic use and duration, PCT has been evaluated for antibiotic initiation, antibiotic cessation, and the combination of both strategies. These strategies assume PCT availability from an institutional laboratory. If the PCT turnaround time is more than 24 hours, the effects of minimizing antimicrobial treatment days may be limited.
   5. It is unclear how to interpret PCT levels in certain patient populations; however, PCT use is not precluded in these populations as long as providers are aware of limitations. According to current evidence, PCT should be used with caution in the following patient populations:
      a. Chronic kidney disease: Patients with chronic kidney disease have higher baseline levels of procalcitonin.
      b. PCT levels can be elevated among patients during times of noninfectious conditions such as trauma, surgery, burns, immunomodulator therapy that increases proinflammatory cytokines, cardiogenic shock, during peritoneal dialysis, and cirrhotic patients.
   6. According to current evidence, PCT should not routinely be measured in patients without signs and symptoms of infection. The decision to initiate patients on antibiotics without signs and symptoms of infection using PCT alone would probably lead to antimicrobial overuse and possible adverse effects associated with antimicrobial therapy. In a PCT study of critically ill patients, 1200 patients were randomized to either a PCT alert strategy or a standard of care. For those randomized to intervention, a PCT concentration greater than 1.0 mcg/L generated an alert that mandated clinical intervention, which included microbiological cultures, additional radiologic assessment, and/or initiation or expansion of antimicrobial coverage. Overall, this strategy did not lead to an improvement in mortality or time to appropriate antibiotics. In contrast, patients experienced a greater need for mechanical ventilation, prolonged ICU LOS, and prolonged antibiotic use.
   7. In critically ill patients with signs and symptoms of infection, a baseline PCT (at the time of the symptoms) should not be used to determine whether antibiotics should be initiated. The compliance rate for withholding antibiotics for a low PCT in this scenario has consistently been low. The compliance rate in clinical practice is likely even lower than that in clinical studies; however, this has not been evaluated. If a baseline PCT is obtained, it should be used to trend the PCT for the possible early discontinuation of antibiotics. In a study of patients with signs and symptoms of infections to determine whether a PCT-guided strategy would limit the initiation of antibiotics, no difference in antibiotic use was seen. However, this was probably because only 36% of clinicians were compliant with the recommendation to withhold antimicrobials when the PCT was low. This is in stark contrast with the 86% compliance rate with the recommendation to initiate antibiotics when the PCT was high.
   8. Critically ill patients with signs and symptoms of infection should have a baseline PCT obtained for trending purposes. A low PCT (or substantial decrease from baseline) during antibiotic treatment should be used to shorten the duration of antimicrobial therapy. This could be accomplished through either eliminating unnecessary antibiotics in patients who are not infected or shortening the course of therapy for patients who are infected. This strategy has been proved safe and effective in a wide spectrum of critically ill patients. Several studies have evaluated the utility of a PCT-guided strategy

for determining the appropriate time to discontinue and/or de-escalate antibiotics. These studies consistently show that PCT guidance for discontinuing antimicrobial therapy led to decreases in antibiotic use without an untoward outcome effect. This has been shown in various ICU populations, in patients with different severity of illness, and in those with proven infections. The largest PCT study of critically ill patients (n=1575) was published in 2016. It demonstrated a significant decrease in the use of antimicrobials (median 5 days vs. 7 days) and 28-day mortality (20% vs. 25%; p=0.012). A recent meta-analysis showed that use of PCT to guide discontinuation of antimicrobial therapy may have a short-term mortality benefit.

9. A recent prospective study showed that the inability to decrease PCT by at least 80% by day 4 was an independent predictor of mortality.
10. Many different PCT guidance algorithms exist; Figure 2 represents a reasonable approach to using PCT for antibiotic cessation. In certain scenarios, PCT concentrations may be affected. For example, some clinicians recommend using different cutoffs (e.g., greater than 0.5 mcg/L) in patients with stage 5 chronic kidney disease or requiring dialysis. Furthermore, in some studies, the PCT cutoff is greater than 1.0 mcg/L among patients with recent surgery.
11. Procalcitonin levels have been elevated in patients with COVID-19, although interpretation is challenging as this could represent bacterial co-infection, could be a marker of severity of acute respiratory distress syndrome, or respiratory failure could cause immune dysregulation that increases the production of cytokines. Due to the unclear cause of elevated PCT in COVID-19, it is not advisable to use the PCT level by itself to guide antibiotic therapy.

F. MeMed BV
   1. A host response diagnostic that differentiates bacterial from viral infection at the point of need by computationally integrating the levels of three host immune response proteins (tumor necrosis factor-related, apoptosis-inducing ligand [TRAIL], interferon gamma inducible protein-10 [IP-10], and C-reactive protein [CRP]) into a score indicating the likelihood of a bacterial immune response or co-infection versus a viral infection.
      a. Score: 0–34 = viral infection, 35–65 = equivocal, 66–100 = bacterial infection
      b. Delivers results in 15 minutes
   2. Additional data is needed before wide spread clinical implementation

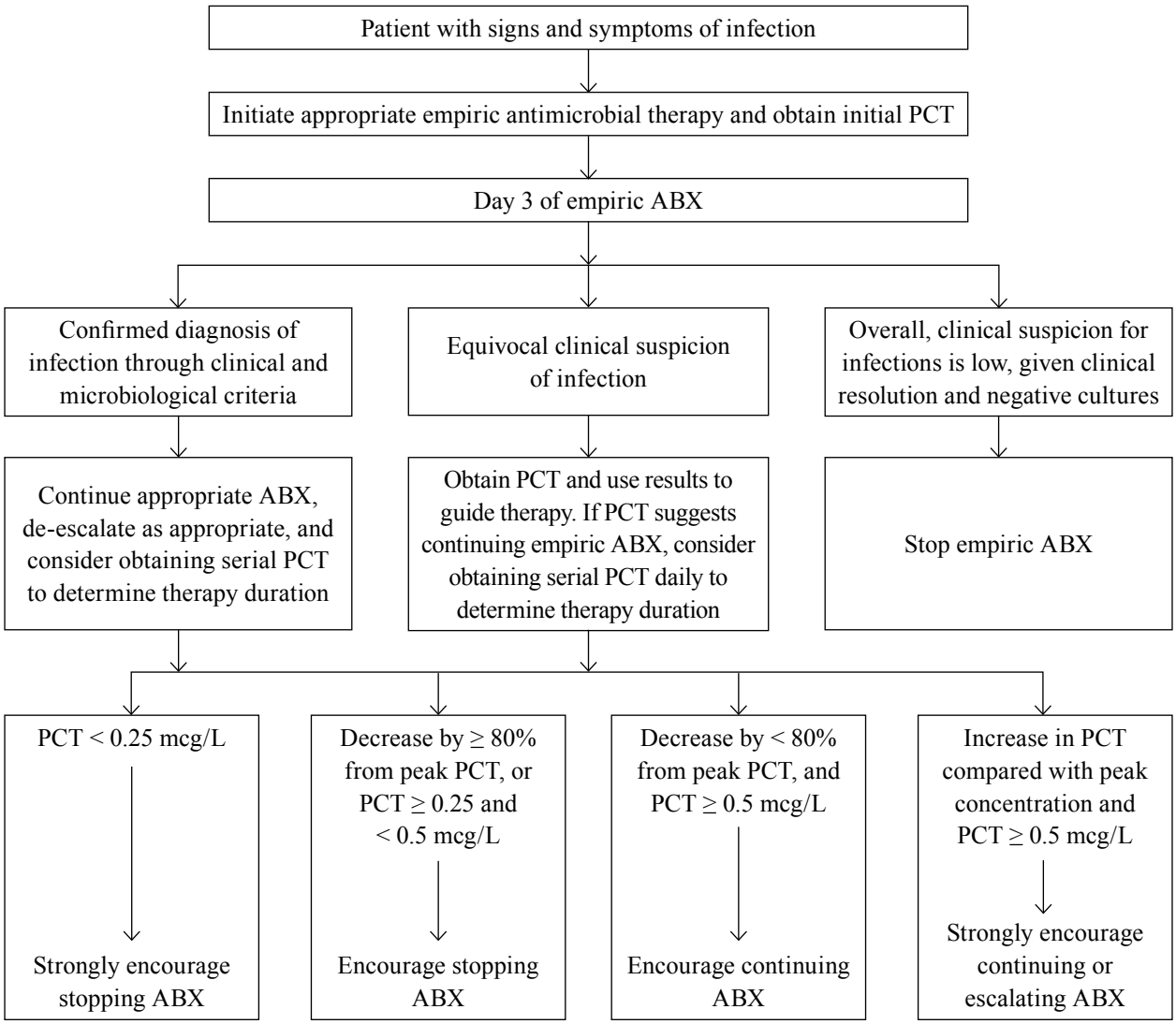


Figure 2. Sample procalcitonin-guided antimicrobial management algorithm.

ABX = antibiotics; PCT = procalcitonin.

Patient Case

4. A 64-year-old woman is admitted to the medical ICU with possible community-acquired pneumonia. The patient is initiated on ceftriaxone and azithromycin. Chest radiography reveals focal infiltrate. On admission, she is dyspneic with a respiratory rate of 33 breaths/minute. Her vital signs and laboratory values are as follows: blood pressure 90/50 mm Hg, heart rate 101 beats/minute, WBC 18 x 10^3 cells/mm^3, and lactate 4.2 mmol/L. A PCT is obtained on admission. The results are available 12 hours after antibiotics are initiated. The PCT result is 0.1 mcg/L. Which is the most appropriate option, given the PCT result?
   A. Continue all current antibiotics.
   B. Discontinue all antibiotics.
   C. Escalate antibiotics to piperacillin/tazobactam.
   D. Discontinue ceftriaxone only.

V. INTERPRETING SUSCEPTIBILITY REPORTS

A. Rationale: Identifying microorganisms and antibiotic susceptibilities is integral to the care of critically ill patients with infections. An understanding of the microbiology laboratory methods for pathogen identification and susceptibility testing will further equip critical care pharmacists with the ability to use the most appropriate antimicrobial regimens for the treatment of critical care–associated infections. Standards for antimicrobial identification, sensitivity testing, and determination of MIC breakpoints are determined by the Clinical & Laboratory Standards Institute (CLSI). CLSI is a volunteer-driven standards development organization that promotes the development and use of laboratory consensus standards and guidelines within the healthcare community.

B. Methods for Antimicrobial Susceptibility Testing (AST)
   1. Disk diffusion (Kirby-Bauer)
      a. Test agar plates are swabbed with a standardized concentration of the test organisms; paper disks containing a defined antibiotic concentration are then placed on the plates. The diameter of the zone of inhibited growth is inversely proportional to the MIC value.
      b. Usually reported qualitatively as sensitive, intermediate, or resistant
   2. Dilution methods (i.e., broth dilution, agar dilution)
      a. Two-fold serial dilutions of antibiotics are prepared (in either broth or agar), which are then inoculated with a standardized concentration of test organisms.
      b. The MIC is determined.
      c. Precision of this method is within one 2-fold concentration in either direction.
      d. Many prefabricated broth microdilution plates are commercially available.
   3. E-test
      a. A calibrated plastic strip with a range of continuous MIC values (with fifteen 2-fold dilutions) is placed on an inoculated agar plate. The MIC can be determined by the intersection between the zone of inhibition and the edge of the calibrated strip.
      b. Complement other systems to determine the MIC for specific antibiotics and test for resistance and for confirmatory testing

c. Studies have shown that the reading of the exact MIC on an E-test strip is subject to the user's interpretation and could lead to a one- or two-dilution error in either direction compared with standardized methods.

4. Automated systems (e.g., Vitek, Microscan, Sensititre, Phoenix, Accelerate Pheno test)
   a. Uses computerized algorithms for interpreting MIC values
   b. Can usually perform AST more quickly than traditional methods
   c. More than 80% of all clinical microbiology laboratories report using an automated system as their primary method of susceptibility testing.
   d. May be unable to detect certain resistance mechanisms (i.e., inducible enzymes)
   e. Standard inoculums of pathogens are used. For infections in which the in vivo inoculum may be higher, certain antimicrobials may have higher MIC values.
5. Specific confirmatory tests for antimicrobial resistance
   a. Macrolide-lincosamide-streptogramin resistance
      i. Strains of *Staphylococcus* spp. can have a transferable resistance mechanism called macrolide-lincosamide-streptogramin, which is inducible by clindamycin and can lead to treatment failure.
      ii. Inducible resistance is not detected by routine AST.
      iii. Detected using double-disk diffusion
   b. ESBL
      i. CLSI procedures exist for *K. pneumoniae*, *Klebsiella oxytoca*, *E. coli*, and *Proteus mirabilis*.
      ii. Initial screening with susceptibility testing for ceftazidime, aztreonam, cefotaxime, or ceftriaxone
      iii. Confirmatory testing with broth microdilution, disk diffusion, or E-test strips. Presence of ESBL is confirmed if adding clavulanic acid results in three 2-fold dilution decreases in the paired-cephalosporin MIC.
      iv. Beginning in 2010, CLSI adopted lower MIC breakpoints for many cephalosporins for Enterobacterales (see Table 7). The rationale for this change is that the lower MIC may more readily detect the presence of ESBL, hence eliminating the need for the labor-intensive confirmatory tests. Compliance with this new practice is variable among clinical microbiology laboratories. Clinicians should inquire within their local clinical microbiology laboratory regarding whether the new standards have been adopted. If a microbiology laboratory has adopted the new CLSI recommendations for ESBL detection, confirmatory tests are unnecessary.

Table 7. Changes to CLSI Enterobacterales Breakpoints with 2020 Updates

Agent	Pre-2010 CLSI Breakpoints			2020 CLSI Breakpoints		
	S	I	R	S	I	R
Cefazolin[a]	≤ 8	16	≥ 32	≤ 2	4	≥ 8
Cefazolin[b]	≤ 8	16	≥ 32	≤ 16	—	≥ 32
Cefotaxime	≤ 8	16–32	≥ 64	≤ 1	2	≥ 4
Ceftriaxone	≤ 8	16–32	≥ 64	≤ 1	2	≥ 4
Ceftazidime	≤ 8	16	≥ 32	≤ 4	8	≥ 16
Aztreonam	≤ 8	16	≥ 32	≤ 4	8	≥ 16

[a]Cefazolin used as therapy for infections other than uncomplicated UTIs.

[b]Cefazolin used as therapy for uncomplicated UTIs.

CLSI = Clinical & Laboratory Standards Institute; UTI = urinary tract infection.

c. AmpC β-Lactamase
   i. Inhibitor-resistant β-lactamase (e.g., clavulanic acid, tazobactam, avibactam, vaborbactam, durlobactam, and relebactam)
   ii. Maintains sensitivity with cephamycins (e.g., cefotetan, cefoxitin) and cloxacillin
   iii. Confirmatory test with E-test containing cephamycin alone or a combination of cephamycin and cloxacillin. If the MIC is decreased by more than three 2-fold dilutions, the presence of AmpC β-lactamase is confirmed.

d. Carbapenemase
   i. Metallo-β-lactamase (e.g., New Delhi metallo-β-lactamase [NDM]; Verona integrin-encoded metallo-β-lactamase [VIM]); imipenimase-type carbapenemase)
      (a) Hydrolyzes both β-lactam and carbapenem antibiotics, but aztreonam maintains susceptibility. Clinically, about 80% of metallo-β-lactamase–containing pathogens are resistant to aztreonam because of other resistance mechanisms.
      (b) Zinc-mediated metallo-β-lactamase can be repressed by ethylene-diamine tetraacetic acid (EDTA). Reduction in imipenem MIC by more than three 2-fold dilutions in the presence of EDTA confirms presence of zinc-mediated metallo-β-lactamase.
   ii. KPC
      (a) KPC enzyme can produce a slightly higher MIC that may still be in the range considered sensitive. This is in contrast to other carbapenem-resistant mechanisms in *Pseudomonas* and *Acinetobacter*, which generally produce a fully resistant MIC. Hence, a carbapenem MIC of 1 mcg/mL or greater in an Enterobacterales should be evaluated through further confirmatory testing.
      (b) Modified Hodge test should be performed for pathogens with an elevated carbapenem MIC.
         (1) Place standard sensitive *E. coli* strain on entire plate.
         (2) Place meropenem or carbapenem disk in center of test area.
         (3) Streak test organism in a straight line from edge of carbapenem disk to edge of plate (up to four different organisms can be tested).
         (4) Positive carbapenemase results from modified Hodge test have a clover leaf–like indentation of the *E. coli* growing along the test organism's streak.
      (c) Starting in 2010, CLSI lowered the susceptibility breakpoints for carbapenem (0.5 mcg/mL and 1.0 mcg/mL or less for ertapenem and other carbapenems, respectively), with the intent of eliminating the need for confirmatory tests.
         (1) Clinicians should inquire about the practices of their local microbiology laboratory regarding carbapenem susceptibility.
         (2) Particularly when confirmatory tests are not being performed, and a laboratory continues to use the older MIC breakpoints, a higher clinical suspicion for carbapenemase is warranted.
         (3) Ertapenem resistance seen on AST is a sensitive marker for the presence of carbapenemase. Before the 2010 reclassification of carbapenem susceptibility breakpoints, there were reports of up to 46% of clinical isolates with genotypic evidence of KPC-producing enzymes being inadvertently labeled as imipenem sensitive.
         (4) Deleterious outcomes associated with the use of carbapenems in carbapenemase-producing organisms have been reported.

C. MIC Breakpoints: Determining the correct MIC breakpoint by the governing standards is a complicated process. CLSI promotes the development of voluntary consensus standards within the medical community. Determining CLSI breakpoints could be based on several conflicting interests, which may include some of the following:
   1. Microbiological: MIC that distinguishes wild-type bacteria from those that have acquired additional resistance mechanisms
   2. PK/PD: Derived from human or animal data and modeled to determine the likelihood that standard prescribed doses will meet specified PD criteria for suppressing bacterial growth
   3. Breakpoints that may decrease the need for confirmatory tests, hence decreasing the microbiology laboratory workload. Such an approach, although possibly sensitive for determining the presence of resistance mechanisms, may potentially lead to the compensatory use of broader antimicrobials.
   4. The CLSI breakpoints may be inconsistent with the breakpoints established by the FDA. Automated systems report MIC breakpoints according to FDA approvals. Changes to reporting and labeling require additional clearance from the FDA. Unfortunately, the FDA labeling may not be up to date with the current CLSI recommendations. Hence, despite newer CLSI recommendations, many institutions may still be reporting susceptibility results that are based on their respective FDA labeling breakpoints. These breakpoints may result in pathogens being labeled as sensitive, even though, according to the current CLSI standards, they would be considered resistant.
   5. The exact MIC breakpoints that are used by a local microbiology laboratory can be variable, depending on the laboratory's adaptation of new standards. In addition, there are significant variations in the FDA-approved breakpoints used by automated AST methods and updated CLSI guidelines. Once again, this highlights the importance of a clinician's understanding of the exact methods and breakpoints that are used in his or her institution.

D. Minimum recommended information that critical care pharmacists should recognize about their local microbiology laboratory practices
   1. Routine methods for AST
   2. Availability of rapid diagnostic tests for speciation or resistance gene identification
   3. If additional tests are available (E-test, genotyping, etc.) and how to go about requesting additional information
   4. Are confirmatory tests for resistance mechanisms routinely done (e.g., double-disk diffusion for macrolide-lincosamide-streptogramin–inducible resistance, ESBL and KPC confirmatory tests)?
   5. What is the usual procedure for censoring the AST results?
      a. Does the AST result try to guide clinical decision (i.e., if pathogen were sensitive to oxacillin, would vancomycin results be censored)?
      b. If certain resistance mechanisms are detected, would the laboratory automatically update susceptibility results (i.e., if ESBL were detected, would β-lactam/β-lactamase combination antibiotics automatically be changed to resistant)?
      c. Are any antibiotics part of the standard panel but routinely hidden from clinical reports?
      d. Most microbiology laboratories would retain all the information, even if parts of the results had been censored. Critical care pharmacists should consider contacting the microbiology laboratories to see whether additional information regarding the pathogen is available.
   6. When an MIC is reported, is only the MIC breakpoint reported (i.e., MIC 2 mcg/mL or less), or is the actual MIC reported (i.e., 0.25 mcg/mL)?
   7. Are the breakpoints reported consistent with the updated CLSI recommendations?

E. Clinical Application of Interpreting AST
   1. The ideal application of AST interpretation and clinical therapeutics may include the following:
      a. Knowledge of the individual pathogen and the corresponding MIC breakpoints for each of the tested antibiotics
      b. Potential reasoning behind each of the clinical breakpoint designations (i.e., wild-type selection, PK/PD breakpoints assessment, screening for resistance)
      c. Accurate comprehension of the local AST methods used and potential limitations of the MIC reported
      d. Understanding of the tested antimicrobial's PD parameters, which conveys the highest likelihood for eradication of the microorganism
      e. Selecting the most appropriate antibiotic according to an assessment of the following:
         i. What genotypic resistance mechanisms may be suggested by the phenotypic representation of the AST?
         ii. How may the underlying resistance mechanisms affect the choice of antimicrobial therapy?
         iii. The PK characteristics of the medication and patient characteristics that may affect the antimicrobial's PK
         iv. The likelihood that the considered antimicrobial agent will reach its PD targets for optimal therapy at the local site of infection
         v. Selecting the antimicrobial according to a balance of antimicrobial stewardship principles, adverse effects profile, allergies, and cost
   2. When an antimicrobial must be chosen despite its having a higher-than-ideal MIC, the penetration into the site of infection is questionable, or the correlation of PK and PD target attainment is not available, additional interventions may be required. Optimization of therapy should be tried through one or more of the following: using a higher dose, using administration strategies that may optimize PK/PD (i.e., extended- or continuous-infusion β-lactams), or combination therapy.

VI. MECHANISMS OF ANTIBACTERIAL RESISTANCE AND TREATMENT OF MULTIDRUG-RESISTANT PATHOGENS

A. Epidemiology and Clinical Significance: The WHO has identified antimicrobial resistance as one of the three greatest threats to human health.
   1. The National Healthcare Safety Network for the CDC reports recent resistance rates among commonly encountered pathogens. More than 3000 U.S. hospitals participate in this national surveillance program, with as many as 100,000–300,000 pathogens reported and about 70% of the reporting from critical care sites. See Table 8 for selected resistance rates in different hospital-acquired infections.
   2. Resistance rates continue to be high, with trends toward an increase in certain pathogens compared with the reports from previous years.
   3. Resistant pathogens consistently correlate with worse clinical outcomes, which is partially explained by the higher likelihood of empiric treatment with a resistant antibiotic.
   4. The antibiotic pipeline has slowed down considerably, with consistent decreases in FDA approvals of antimicrobial agents during the past 3 decades. The combination of prevailing resistance, including the emergence of pan-resistant pathogens, with the lack of new antimicrobial agents presents a potential global health problem.

5. An understanding of resistance mechanisms would assist the ICU clinician in effectively treating current resistant pathogens while incorporating antimicrobial stewardship principles to prevent further resistance. See Table 9 for common mechanisms of resistance.

Table 8. Recent Resistance Rates in Hospital-Acquired Infection (2015–2017)

Pathogen	CLABSI % Resistance	CAUTI % Resistance	VAP % Resistance	SSI % Resistance
Gram-negative Bacteria				
E. coli	n=1129	n=14,390	n=520	n=15,302
Third-generation cephalosporins	29.8	16.0	23.8	18.5
Carbapenem	2.4	0.3	1.0	0.6
Fluoroquinolone	43.2	32.7	32.7	34.7
Multidrug resistant[a]	14.4	6.9	11.8	7.8
K. pneumoniae/oxytoca	n=1708	n=5775	n=936	n=4591
Third-generation cephalosporins	29.6	16.6	14.3	14.3
Carbapenem	11.0	5.1	5.4	3.5
Multidrug resistant[a]	20.2	9.6	8.7	6.4
P. aeruginosa	n=1061	n=5047	n=1192	n=4405
Aminoglycosides	14.9	15.3	13.6	6.8
Ceftazidime or cefepime	26.5	18.6	25.9	12.3
Fluoroquinolones	27.1	24.9	26.7	12.3
Carbapenem	26.3	19.7	26.3	11.2
Piperacillin/tazobactam	20.2	13.8	21.7	9.6
Multidrug resistant[a]	18.6	13.6	16.8	5.5
Enterobacter spp.	n=1078	n=2120	n=781	n=3471
Third-generation cephalosporins	12.2	12.6	6.7	7.3
Carbapenem	7.2	6.5	3.5	3.7
Multidrug resistant[a]	7.7	6.2	0.8	2.6
A. baumannii	n=392	n=197	n=294	n=118
Carbapenem	47.2	52.8	41.3	32.3
Multidrug resistant	46.6	53.1	37.5	33.6
Gram-positive Bacteria				
S. aureus	n=2497	n=655	n=2673	n=5909
Oxacillin	50.0	40.5	36.9	55.8
E. faecalis	n=2117	n=3881	NR	n=7584
Vancomycin	8.5	4.2		3.4
E. faecium	n=1981	n=1034	NR	n=3584
Vancomycin	89.5	82.8		53.1

[a]*Multidrug resistant* is defined as being either intermediate or resistant to at least one drug in three of the following five classes: third- and fourth-generation cephalosporins, fluoroquinolones, aminoglycosides, carbapenems, and piperacillin and/or piperacillin/tazobactam (definition according to CDC National Healthcare Safety Network).

CAUTI = catheter-associated urinary tract infection; CLABSI = central line–associated bloodstream infection; NR = not reported; SSI = surgical site infection; VAP = ventilator-associated pneumonia.

Table 9. Common Resistance Mechanisms

Antibacterial Agent	Mechanism of Resistance
β-Lactams	β-Lactamases (AmpC, ESBL, KPC) Reduced permeability Altered penicillin binding proteins Increased efflux
Fluoroquinolones	Altered DNA gyrase and topoisomerase Increased efflux Decreased protein targets
Macrolides	Increased efflux Methylating enzymes
Aminoglycosides	Aminoglycoside-modifying enzymes Increased efflux Modification of target proteins
Glycopeptides	Altered target Decreased permeability
Tetracyclines	Increased efflux Protection of target proteins
Trimethoprim	Increased efflux Altered dihydrofolate reductase
Rifamycin	Altered RNA polymerase

B. Mechanisms of Resistance (bacteria often possess several mechanisms)
   1. Decreased permeability (i.e., porin loss, thickened cell wall)
   2. Increased efflux (i.e., macrolide efflux pump)
   3. Target modification (i.e., alteration in penicillin-binding proteins)
   4. Hydrolysis (i.e., β-lactamases, aminoglycoside-modifying enzymes)

C. Factors Associated with Resistance Acquisition
   1. Crowding of patients with high levels of disease acuity and/or antimicrobial use
   2. Prolonged hospital LOS
   3. Colonization pressure: Proportions of people colonized with resistant bacteria. Combated by strict compliance with infection control procedures to prevent colonization, adequate nurse staffing ratios, and hand hygiene
   4. Use of invasive devices (endotracheal tubes, intravascular catheters, and urinary catheters)
   5. Previous or prolonged use of antibiotics

D. Clinical Approach to Common Pathogen Resistance Seen in Critically Ill Patients
   1. ESBL
      a. Confers resistance to third-generation cephalosporins and aztreonam
      b. Found primarily in *E. coli* and *K. pneumonia*e spp. but can also be seen in other Enterobacterales.
      c. Carbapenems should be considered the drug of choice in severe infections.
      d. Non–β-lactams could be used if they showed sensitivity on AST; however, because ESBLs are usually plasmid mediated, there are often other acquired resistance mechanisms. The rates of cross-resistance to other classes of antibiotics are 55%–100%.

e. β-lactam/β-lactamase inhibitors and cefepime often have in vitro activity, though clinical failures have been reported.
   i. A previous post hoc analysis found that use of β-lactam/β-lactamase inhibitors was not associated with worse outcomes for ESBL-producing *E. coli* bacteremia compared with carbapenems.
   ii. However, a recent prospective randomized controlled noninferiority study evaluating definitive treatment of ceftriaxone-resistant *E. coli* or *K. pneumoniae* found increased mortality with piperacillin/tazobactam compared with meropenem (12.3% vs. 3.7%). These results do not support the use of piperacillin/tazobactam in the treatment of ESBL-producing organisms.
   iii. Data are conflicting regarding the use of cefepime for treating ESBL infections. Some studies show worse outcomes, whereas others show no difference compared with carbapenems. This may partly be explained by the cefepime MIC distribution. Traditionally (before 2014), the susceptibility breakpoint for cefepime for Enterobacterales was 8 mcg/mL or less. However, in 2014, CLSI recommended decreasing the sensitive cefepime MIC breakpoint to 2 mcg/mL. In addition, a new category for sensitive dose-dependent was created, where maximal doses of cefepime are recommended for MICs of 4 and 8 mcg/mL. Hence, for many years, cefepime may have been used in ESBL infections when the MIC was 4–8 mcg/mL, but doses were not optimized.
   iv. Of interest, a recent investigation correlated cefepime MIC to ESBL Enterobacterales with mortality, where a cefepime MIC of 1 mcg/mL or less was associated with significantly lower mortality compared with higher MIC values. Together, given the conflicting clinical results, it is difficult to endorse the use of cefepime for the treatment of ESBL infections. However, in a stable patient with an ESBL infection having a cefepime MIC of 1 mcg/mL or less, cefepime may be considered for consolidative therapy to minimize carbapenem use.

2. AmpC β-lactamases
   a. Confer resistance to penicillins and narrow-spectrum cephalosporins
   b. β-Lactamase inhibitor resistant; hence, tazobactam and clavulanic acid would not provide additional coverage
   c. Innate low-level production in many gram-negative bacteria (i.e., *Enterobacter*, *Citrobacter*, *Serratia*, *Morganella*, *Providencia*, and *Pseudomonas*). This low-level production leads to resistance against penicillin, ampicillin, and first-generation cephalosporins.
   d. Genetic mutation leading to sustained high-level production is possible (derepressed mutants).
      i. Theoretically, treatment with third-generation cephalosporins or broad-spectrum cephalosporins for non-derepressed mutants may select species with the mutation. The actual incidence of breakthrough infections is unknown with derepressed mutants when bacteria with innate AmpC β-lactamase production are treated with third-generation cephalosporins or extended-spectrum penicillins. However, it is believed to be low.
      ii. Several epidemiologic studies have linked the use of third-generation cephalosporins with the increase in pathogenic species of AmpC hyperproducing Enterobacterales.
      iii. Derepression confers resistance to third-generation cephalosporins and broad-spectrum penicillins (piperacillin, ticarcillin).
      iv. Commonly occurs in *E. cloacae* and *Citrobacter freundii,* Hafnia alvei, Klebsiella (Enterobacter) aerogenes, Yersinia enterocolitica
      v. Cefepime generally retains activity against derepressed mutants.
      vi. Carbapenem and cefepime should be considered the drugs of choice in severe infections.
      vii. Because derepression results from the mutation of a chromosomal-mediated β-lactamase, and not from transmitted plasmids, many pathogens with this mutation maintain sensitivity to other nonβ-lactam antimicrobials (e.g., fluoroquinolones, aminoglycosides).

Patient Case

5. A 74-year-old man is admitted to the surgical ICU after elective hip replacement surgery. The patient, who has a history of chronic pulmonary obstructive disease, cannot be weaned off the ventilator after surgery. During the patient's course, he develops signs and symptoms of infection. His vital signs and laboratory values are as follows: blood pressure 94/55 mm Hg, heart rate 114 beats/minute, temperature 101.9°F (38.8°C), WBC 18 x 10^3 cells/mm^3, and lactate 3.2 mmol/L. The patient is empirically initiated on piperacillin/tazobactam and vancomycin and given 2 L of crystalloid fluids; pan cultures are sent. Urinalysis reveals pyuria, positive leukocyte esterases, and nitrites. Blood and sputum cultures are negative, but urine culture shows *E. coli*. The patient's urinary catheter is removed, and vancomycin is discontinued. On day 3 of therapy, antibiotic susceptibility results are available. The patient's *E. coli* is resistant to third-generation cephalosporins with laboratory confirmation of the presence of ESBL. The laboratory reports the following antimicrobials and corresponding MIC values: piperacillin/tazobactam less than 2 mcg/mL – S; cefepime 4 mcg/mL – SDD; imipenem 0.5 mcg/mL – S; and ciprofloxacin 0.25 mcg/mL – S. The patient's vital signs and laboratory values are as follows: blood pressure 110/70 mm Hg, heart rate 98 beats/minute, respiratory rate 30 breaths/minute, temperature 98.7°F (37.1°C), and WBC 9 x 10^3 cells/mm^3. Which is the most appropriate antibiotic option?
   A. Change piperacillin/tazobactam to imipenem.
   B. Continue piperacillin/tazobactam alone.
   C. Change piperacillin/tazobactam to cefepime.
   D. Add ciprofloxacin to piperacillin/tazobactam.

3. Carbapenemase
   a. Seen in *Acinetobacter*, *Pseudomonas*, and Enterobacterales
   b. The global spread of carbapenem resistance has become an epidemic.
   c. The CDC continues to report that carbapenem-resistant Enterobacterales (CRE) is at an urgent hazard level, where high consequence and probability for widespread public health concerns exist.
   d. Confers resistance to most β-lactams, including carbapenems, cephalosporins, monobactam, and broad-spectrum penicillins
   e. Treatment options
      i. Tigecycline:
         (a) Glycylcycline antibiotic, which is structurally similar to tetracyclines
         (b) Mechanism of action: Inhibition of 30s ribosomal subunit
         (c) Spectrum of activity:
            (1) Gram-positive bacteria: *Enterococcus* (including vancomycin-resistant enterococci), *Listeria*, *Staphylococcus* (including MRSA/CoNS), *Streptococcus*
            (2) Most gram-negative bacteria, including *Acinetobacter*, ESBL-producing Enterobacterales, derepressed AmpC Enterobacterales, CRE, and *Stenotrophomonas*
            (3) Anaerobes, including *Bacteroides* and *Clostridium*
            (4) Atypicals
            (5) Does not cover *Pseudomonas*, *Providencia*, *Proteus, or Morganella*
         (d) PK
            (1) Wide volume of distribution: 7–10 L/kg
            (2) The intracellular distribution of tigecycline results in a decreased serum/tissue concentration ratio. This has led many clinicians to state that tigecycline is not the ideal drug for bloodstream infections. However, tigecycline has not been evaluated exclusively for the treatment of bloodstream infections. In a pooled analysis of eight

studies, patients with secondary bacteremia treated with tigecycline were compared with patients treated with other antibiotics. Overall, no significant differences in outcomes were seen. Despite these results, clinicians should exert caution with the use of tigecycline for bacteremia and should reserve tigecycline for pathogens when no other antibiotics are viable options.

(3) Primarily biliary elimination

(4) Urinary elimination is 8%–11%. The low urinary elimination limits tigecycline's role in treating UTIs. The use of tigecycline for treating multidrug-resistant UTIs has been reported only in case reports, with most reports showing treatment success. However, a recent evaluation of tigecycline for the treatment of KPC bacteriuria indicated a correlation with the subsequent development of tigecycline resistance. Without further data, tigecycline should not routinely be used for UTIs when other treatment options are available.

(5) Poor penetration into lung epithelial lining fluid, which is in contrast to high penetration into lung alveolar cells. This characteristic may partly explain the findings of a study evaluating tigecycline compared with imipenem for the treatment of hospital-acquired pneumonia. This study showed that tigecycline had a lower treatment success rate than imipenem. The difference in treatment success was mainly attributable to the significant differences in patients with ventilator-associated pneumonia. There was also a trend toward increased mortality in the subgroup of patients with ventilator-associated pneumonia.

(e) The FDA issued a safety warning in 2010 indicating a possible increased mortality risk associated with the use of tigecycline compared with other drugs used to treat a variety of other serious infections. This was compiled using several phase III studies in which tigecycline had been proven noninferior to other standard treatments. Subsequently, several other meta-analyses were published with conflicting results regarding tigecycline's increased mortality risk.

(f) Given tigecycline's possible shortcomings, it seems prudent to avoid the routine use of tigecycline in infections when other treatment options are available. However, tigecycline may be one of the few remaining antibiotic options for treating carbapenem-resistant pathogens. When tigecycline must be used because of limited treatment options, clinicians should consider administering combination therapy with other agents that have in vitro susceptibility.

(g) Resistance to tigecycline has been reported. Clinicians should seek confirmation from their microbiology laboratory regarding tigecycline sensitivity. Of note, CLSI currently has no recommendation for MIC breakpoint for tigecycline against *Acinetobacter* spp. The MIC breakpoint for sensitivity against Enterobacterales is 2 mcg/mL or less.

ii. Polymyxins:

(a) Colistin (polymyxin E)

(1) Mechanism of action: A cationic cyclic decapeptide that functions by displacing calcium and magnesium from the outer cell membrane, hence changing the permeability of the cell membrane to allow insertion of the molecule into the cell membrane. Once the molecule is inserted into the cell membrane, it disrupts the cell membrane integrity and subsequently leads to cell death.

(2) Spectrum of activity: Covers only gram-negative bacteria, including CRE. Does not cover *Proteus*, *Providencia*, *Burkholderia*, *Serratia*, or *Stenotrophomonas*

(3) First used in the United States in the 1960s but fell out of favor because of reports of nephrotoxicity and neurotoxicity

(4) Around 2000, with the emergence of CRE, colistin use was reconsidered, given the lack of treatment options.
(5) Dosing challenges
  (A) Different products provide different dosing recommendations.
  (B) Colomycin injection is the brand primarily used in Europe. Dosed in international units (IU).
  (C) Coly-Mycin is the brand primarily used in the United States. Dosing in colistin-based activity
  (D) 3 million IU of colistin is equal to about 100 mg of colistin-based activity.
(6) PK/PD
  (A) All colistin products, regardless of the dosing units, are administered as colistimethate, which is a prodrug. Colistimethate is hydrolyzed to the active drug colistin.
  (B) Colistimethate is excreted through renal clearance, whereas the active drug colistin is cleared by nonrenal pathways. Hence, renal dysfunction leads to a higher portion of colistimethate being present for hydrolysis into colistin, thus increasing the final active drug concentration.
  (C) PD parameter for maximal activity is area under the curve/MIC ratio.
  (D) In the 1960s, the dosing recommendations and PK/PD parameters were largely unknown because the methods for evaluations were drastically different from the current standards.
  (E) Recent PK/PD evaluations show that traditional dosing methods are probably insufficient to reach adequate serum concentrations to maximize PD target attainment.
  (F) Several studies have reported that a higher colistin dose is associated with significant improvements in clinical outcomes.
  (G) International consensus dosing recommendations
    • 300 mg colistin base activity loading dose.
    • 300-360 mg colistin base activity divided every 12 hours, with adjustments made based on renal function / renal replacement therapy.
  (H) Several ongoing investigations are evaluating the effects of high-dose colistin regimens and the utility of using a loading dose.
    • One study from South America is recruiting patients to evaluate the utility of a 200-mg loading dose.
    • One study reported a 300-mg loading dose regimen, followed by 150 mg every 12 hours. This study was non-comparative, but it reported high rates of clinical success.
(7) Because the PK/PD of colistin is unclear and little is known regarding the optimal dosing regimen, colistin should be reserved for infections in which other treatment options are not available. In such cases, it may be prudent to administer combination therapy. Furthermore, many in vitro studies have shown synergy between colistin and other antimicrobials.
(8) Resistance to colistin has been reported. In vitro studies also show that colistin resistance develops quickly, which may be another rationale for providing combination therapy. Clinicians should verify colistin susceptibility with their local microbiology laboratory.
  (A) Routine monitoring of colistin concentrations is currently infeasible.
  (B) According to PK data, an MIC of 2 mcg/mL or less should be considered sensitive.

(9) Empiric colistin therapy: In an analysis of the association between covering empiric antibiotics and mortality for infections caused by carbapenem-resistant gram-negative bacteria, the authors concluded that empiric use of colistin before pathogen identification, with or without a carbapenem, was not associated with survival.

(b) Adverse reaction: Nephrotoxicity
  (1) Nephrotoxicity
    (A) 0%–45% reported in recent literature, depending on the definitions used for renal dysfunction
    (B) In general, seems to be dose-dependent
    (C) Usually reversible, with few cases leading to a prolonged need for renal replacement therapy
  (2) Neurotoxicity
    (A) Ranging from paresthesia to apnea
    (B) Incidence of 8%–27% reported in historical studies
    (C) Mentioned only in case reports in the current era of colistin use

(c) Polymyxin B:
  (1) Mechanism of action, spectrum of activity, and adverse reactions similar to those of colistin
  (2) Available in the United States but not in Europe and Australia
  (3) PK/PD
    (A) Administered as an active drug
    (B) PD targets similar to those of colistin
      • Similar unknowns regarding best dosing regimen to achieve PD parameters
      • Manufacturer-recommended dosing: 1.5–2.5 mg/kg/day divided every 12 hours. Recommends renal dose adjustment, but recent studies suggest minimal renal clearance.
    (C) International consensus dosing recommendations
      • Loading dose of 2–2.5 mg/kg total body weight
      • 1.25–1.5 mg/kg total body weight every 12 hours

(d) International consensus guidelines for the optimal use of polymyxins
  (1) Use therapeutic drug monitoring, when possible
  (2) Recommend hospitals have access to both polymyxins
    (A) Polymyxin B preferred for routine systemic use because of superior and more predictable PK
    (B) Colistin preferred agent for UTIs
    (C) Recent systematic review suggests that polymyxin B is associated with less acute kidney injury than colistin.

iii. Ceftazidime/avibactam
  (a) Avibactam is a new β-lactamase inhibitor approved by the FDA in 2015.
  (b) Spectrum of activity: Broad gram-negative activity, including multidrug-resistant *Pseudomonas* and Enterobacterales. Adding avibactam allows coverage against KPC-producing bacteria, together with coverage against other β-lactamases (OXA, CTX-M, AmpC). Minimal coverage against metallo-β-lactamase-expressing strains, *Acinetobacter* spp., gram-positive, and anaerobes.
  (c) FDA approved for the treatment of complicated intra-abdominal infection and complex UTIs. However, most clinicians will reserve its coverage for difficult-to-treat pathogens with minimal coverage options, such as KPC-producing Enterobacterales.

(d) An in vitro study that included 120 KPC-producing pathogens showed good activity with ceftazidime/avibactam.
(e) A recent multicenter-observational, propensity score-weighted study comparing ceftazidime/avibactam with colistin as the initial treatment for CRE showed significant improvements in mortality. These findings require confirmation in a randomized controlled study.
(f) Study evaluating compassionate use of ceftazidime/avibactam as a second-line agent compared with a matched cohort of patients with CRE bacteremia treated with a different agent illustrated significant improvements in mortality (36.5% vs. 55.8%, p=0.005).
(g) No major adverse effects were seen with ceftazidime/avibactam in phase II and phase III studies.
(h) Currently, the average wholesale price for ceftazidime/avibactam is about $1000 per day with normal dosing. This is similar to meropenem/vaborbactam but considerably more expensive than other β-lactams.

iv. Meropenem/vaborbactam
(a) Vaborbactam, which alone has no antimicrobial activity, is a novel cyclic boronic acid-based β-lactamase inhibitor that potentiates the activity of meropenem.
(b) Spectrum of activity: Broad gram-negative activity, including multi-drug resistant Enterobacterales, and remains stable in the presence of ESBLs and AmpC β-lactamases. Vaborbactam is a potent inhibitor of class A carbapenemases, including KPC and class A and C β-lactamases. It is not found to expand coverage to carbapenem-resistant *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, or *Stenotrophomonas maltophilia*.
(c) FDA-approved in 2017 for complicated UTIs. However, most clinicians reserve its coverage for difficult-to-treat pathogens with minimal coverage options, such as KPC-producing Enterobacterales.
(d) A recent multicenter, randomized, prospective, open-label, comparative trial evaluated the efficacy of meropenem/vaborbactam to best available therapy (including mono- or combination therapy with polymyxin B or colistin, carbapenems, aminoglycosides, tigecycline, or ceftazidime/avibactam) for the management of complicated UTI, healthcare-associated pneumonia, ventilator-associated pneumonia, bloodstream infections, and complicated intra-abdominal infections. Meropenem/vaborbactam demonstrated improved efficacy and decreased adverse events compared to best available therapy. Of note, only one patient received ceftazidime/avibactam; therefore, no conclusions can be drawn regarding comparative efficacy between these two agents.

v. Imipenem/cilastatin/relebactam
(a) Cilastatin is a dehydropeptidase-1 inhibitor that prevents inactivation of imipenem by renal dehydropeptidase-1.
(b) Relebactam is a non-β-lactam β-lactamase inhibitor that is active against class A β-lactamases. Compared to avibactam, relebactam has less inhibitory activity against OXA-48, and neither inhibitor is active against metallo-β-lactamases.
(c) FDA-approved in 2019 for complicated UTIs, intra-abdominal infections, and health care-associated/ventilator-associated pneumonia. However, most clinicians reserve its coverage for difficult-to-treat pathogens with minimal coverage options, such as KPC-producing Enterobacterales.
(d) In vitro studies have demonstrated that relebactam, when added to imipenem/cilstatin, restores activity against imipenem nonsusceptible strains that produce KPC.

(e) A recent multicenter, randomized, double-blind, comparator controlled trial compared the efficacy and safety of imipenem/relebactam and colistin combined with imipenem/cilastatin in patients with imipenem-resistant gram-negative bacterial infections (health care-associated/ventilator-associated pneumonia, complicated intra-abdominal infections, or complicated UTI). Clinical response rates were higher for imipenem/relebactam compared to imipenem/colistin.

vi. Cefiderocol
   (a) Novel siderophore cephalosporin with a unique mechanism of action that confers activity against a broad spectrum of resistant gram-negative bacteria, including Enterobacterales, *Pseudomonas*, *Acinetobacter*, and *Stenotrophomonas*. It has no clinically relevant in vitro activity against most gram-positive bacteria and anaerobes.
   (b) Cefiderocol is able to chelate ferric ions and use bacterial iron transport systems, which has been called a "Trojan Horse" strategy and translates to higher concentrations of susceptible gram-negatives in the periplasmic space.
   (c) Due to the cephalosporin/catechol structure, it has increased stability against hydrolysis by clinically relevant β-lactamases, including serine (including KPCs and OXA-48) and metallo-β-lactamases. Given the use of the iron transport pathway, cefiderocol does not require classic β-lactam porin channels to penetrate the cells, and therefore, retains activity in the presence of porin channel mutations and remains unaffected by common efflux pumps present in Enterobacterales.
   (d) FDA-approved in 2019 for complicated UTIs.
   (e) In vitro studies have demonstrated excellent activity of cefiderocol against class A, B, and D CRE.
   (f) A multicenter, randomized, open-label study of cefiderocol compared to best-available therapy for the treatment of severe infections (hospital-acquired pneumonia, ventilator-associated pneumonia, complicated UTI, bloodstream infection, and sepsis) caused by carbapenem-resistant gram-negative pathogens. All-cause mortality versus other antibiotics in critically ill patients was higher in the cefiderocol arm. The cause of higher mortality was not established.

vii. Eravacycline
   (a) Novel fluorocycline of the tetracycline class with a similar structure to tigecycline, escaping many resistance mechanisms seen in other tetracyclines
   (b) FDA-approved in 2018 for complicated intra-abdominal infections. It was investigated for use with UTIs, but outcomes were not favorable.
   (c) Demonstrated in vitro activity against most gram-positive and gram-negative pathogens, including CRE and *Acinetobacter*, and anaerobes. It lacks activity against *Pseudomonas*.
   (d) Low oral bioavailability; therefore, only IV formulation is FDA-approved.
   (e) Randomized, double-blind, multicenter trial found that eravacycline was noninferior to ertapenem for the management of complicated intra-abdominal infections.

viii. Omadacycline
   (a) Omadacycline is an aminomethylcycline antibiotic in the tetracycline class escaping many resistance mechanisms seen in other tetracyclines.
   (b) FDA approved in 2018 for community-acquired bacterial pneumonia and SSIs
   (c) Has activity against most gram-positive and gram-negative pathogens, including ESBL-producing isolates, Acinetobacter, Stenotrophomonas, anaerobes, and atypicals. It lacks activity against Pseudomonas.
   (d) Available as an injectable and oral
   (e) Randomized, double-blind, multicenter trial found that eravacycline was noninferior to moxifloxacin for the management of community-acquired bacterial pneumonia.

ix. Sulbactam/durlobactam
   (a) Only agent developed for the treatment of carbapenem-resistant Acinetobacter baumanii (CRAB)
   (b) Sulbactam has intrinsic activity to Acinetobacter spp., whereas durlobactam had broad-spectrum activity to class A, C, and D β-lactamases.
      (1) Antimicrobial activity of sulbactam is because of its ability to inactivate penicillin-binding protein.
      (2) Durlobactam is a non–β-lactam/β-lactamase inhibitor with expanded coverage of class D β-lactamases.
   (c) FDA approved in 2023 for the treatment of hospital-acquired and ventilator-associated bacterial pneumonia
   (d) A randomized multicenter trial showed sulbactam/durlobactam was noninferior to colistin for the primary efficacy end point of 28-day all-cause mortality for the treatment of CRAB.

x. Combination therapy
   (a) Several retrospective studies suggested that the use of combination therapy is warranted for the treatment of CRE, particularly a regimen involving colistin, tigecycline, and a carbapenem. Interpreting these studies is difficult. Many were noninterventional, hence leading to the evaluation of many treatment regimens. The increased ratio of the number of treatment regimens to the number of patients substantially increases the risk of spurious findings. If no optimal therapy exists for a patient, combination therapy may be considered. These therapies include scenarios in which the carbapenem MIC may be slightly elevated or in which therapies with suboptimal PK/PD (i.e., colistin and tigecycline) are used.
   (b) A recent retrospective study that evaluated the role of combination therapy for carbapenem-resistant gram-negative pathogens showed that combination therapy with several agents, all of which had in vitro sensitivity, led to improvements in outcomes. In contrast, combination therapy with several agents, not all of which had in vitro sensitivity, did not lead to improvements.
   (c) Several case reports recommend considering combining ertapenem with another carbapenem for the treatment of carbapenem-resistant pathogens. This approach takes advantage of the increased affinity for ertapenem seen in vitro with carbapenemases. Hence, administering ertapenem as a sacrificial carbapenem may allow a different carbapenem to exert its effects. Recent systematic review identified 171 patients who were treated with this combination carbapenem strategy, and found clinical and microbiological success reported in 70% of patients. However, this practice requires further testing; hence, it cannot currently be recommended.
   (d) If using polymyxin, combination therapy with another agent that has a susceptible MIC is recommended.

4. Multidrug-resistant *Pseudomonas*
   a. Resistance mechanisms seen with *Pseudomonas aeruginosa* are unpredictable. The presence of several mechanisms, including β-lactamases, porin loss, efflux pump, and alteration of target proteins, complicates treatment options.
   b. Clinical approach usually entails empiric coverage with a β-lactam, which has the best local in vitro activity against *Pseudomonas*, with or without a second antipseudomonal agent. Therapy could be de-escalated to a monotherapy with the narrowest spectrum on availability of AST results.

c. Ceftolozane/tazobactam is a novel β-lactam/β-lactamase inhibitor antimicrobial with enhanced activity against *P. aeruginosa*.
   i. Ceftolozane is a novel fifth-generation cephalosporin.
   ii. Spectrum of activity: Gram-negative organisms, including *P. aeruginosa*. Activity includes coverage against ESBL and AmpC β-lactamase–producing organisms. Limited gram-positive and anaerobic coverage. Among multidrug-resistant and extended drug–resistant *Pseudomonas*, ceftolozane/tazobactam retains good activity, with its MIC_{90} still below the MIC breakpoint for resistant, as determined by the FDA.
   iii. FDA approved for the treatment of complicated intra-abdominal infection, complicated UTI, and hospital-acquired and ventilator-associated pneumonia. However, clinically, its coverage will most likely be reserved for resistant pseudomonal infections. A recent study concluded that high-dose ceftolozane/tazobactam (3 g every 8 hours) was noninferior to meropenem in the treatment of nosocomial pneumonia. Twenty-eight day mortality, clinical response, and adverse reactions were similar between groups.

d. Aminoglycosides
   i. CLSI aminoglycoside breakpoint revisions
      (a) CLSI breakpoints are not recognized by the FDA.
   ii. Recommend tobramycin 7 mg/kg every 24 hours if expanding coverage for Pseudomonas.
      (a) Gentamicin is no longer recommended for Pseudomonas.
      (b) Amikacin may be considered if being used for a urinary source.

5. MRSA
   a. MRSA is a significant cause of both community- and hospital-acquired infections.
   b. Skin and soft tissue infections
      i. Community-acquired MRSA often presents as a skin and soft tissue infection.
      ii. Cutaneous infections and abscesses are best treated with adequate drainage.
      iii. Antibiotics are usually not necessary unless the patient does not respond to drainage, has extensive disease, or has signs and symptoms of systemic infection.
      iv. Antibiotic treatment choices for community-acquired MRSA cutaneous and skin infections include trimethoprim/sulfamethoxazole, clindamycin, and tetracycline.
   c. MRSA bacteremia and endocarditis
      i. Can be treated with either vancomycin or daptomycin
      ii. In a study of patients with right-sided endocarditis, daptomycin was noninferior to vancomycin. Of interest, treatment response was poor with both therapies, with only about 40% in each group meeting the primary outcome of treatment success 42 days after the end of therapy. Daptomycin was dosed at 6 mg/kg in this study.
      iii. Some experts recommend giving higher daptomycin doses (8–10 mg/kg) to optimize the PD target attainment for daptomycin. Daptomycin resistance in *S. aureus* has been reported. A correlation appears to exist between intermediate sensitivity to vancomycin, thought to be caused by a thickened cell wall–limiting penetration, and decreased susceptibility to daptomycin. Currently, the daptomycin breakpoint MIC for MRSA is 1 mcg/mL or less.
      iv. Several investigations have also evaluated daptomycin compared with vancomycin for MRSA bacteremia when the vancomycin MIC was greater than 1 mcg/mL. Although these investigations showed outcome benefit associated with daptomycin use, the studies have several limitations, which may limit their applicability.
         (a) Retrospective analyses
         (b) Daptomycin was usually used as definitive therapy after an initial course of vancomycin.
         (c) Daptomycin use in one study was associated with a significantly higher rate of infectious diseases consultation, which may have other treatment implications.

(d) Given study limitations, the routine use of daptomycin for MRSA infections when the vancomycin MIC is greater than 1 mcg/mL cannot be recommended.
(e) The 2011 IDSA guidelines for the treatment of MRSA state that the treatment of isolates with a vancomycin MIC of 2 mcg/mL or less should be determined by the patient's response to vancomycin, independent of the MIC.

d. MRSA pneumonia
   i. Community-acquired MRSA pneumonia can be treated with either vancomycin or linezolid. Clindamycin can be considered if the strain is susceptible.
   ii. Community-acquired MRSA infections may be associated with the presence of Panton-Valentine leukocidin, which is a two-component staphylococcal membrane toxin that targets leukocytes. This toxin has been linked with severe infections, necrotizing pneumonia, and abscess formation. Theoretically, clindamycin and linezolid, being ribosomal subunit and protein synthesis inhibitors, may attenuate the amount of toxin production.
   iii. A randomized controlled study comparing linezolid to vancomycin included patients with hospital-acquired pneumonia and ventilator-associated pneumonia who had a baseline respiratory culture positive for MRSA.
      (a) Vancomycin was dosed by weight and adjusted locally according to trough levels.
      (b) This was a noninferiority study with nested superiority criteria where the primary end point was clinical response at the end of the study.
      (c) Clinical response was defined as resolution of signs of pneumonia with no further need for other antibiotics.
      (d) Study met both the noninferiority and the superiority criteria with respect to clinical response and microbiological clearance.
      (e) Study may have been confounded by less than 50% of patients reaching vancomycin target concentrations by day 3 and the vancomycin patients having a slightly higher rate of baseline bacteremia.
      (f) No difference in mortality was seen.
      (g) Many clinicians interpret the results from this study as suggesting that either vancomycin or linezolid can be considered for the treatment of MRSA pneumonia.

e. Updates to vancomycin dosing and monitoring
   i. Consensus guidelines for therapeutic monitoring of vancomycin were published in 2020.
   ii. AUC24 of 400–600 mg/hour/L is recommended target to improve clinical efficacy and patient safety. For most isolates, an MIC of 1 mg/L can be assumed.
   iii. Trough targets are no longer recommended for patients with serious MRSA infections.
   iv. Bayesian dosing is the recommended method for dosing adjustments. Bayesian dosing does not require steady-state vancomycin concentrations for early determination of AUC target attainment.

f. Novel agents for the treatment of MRSA
   i. Tedizolid is FDA-approved for the treatment of bacterial skin and skin structure infection.
      (a) Antimicrobial class: Oxazolidinone
      (b) Spectrum of activity: Gram-positive pathogens including MRSA, vancomycin-resistant enterococci, vancomycin-resistant staphylococci, drug-resistant *S. pneumoniae*, and linezolid-resistant gram-positive pathogens
      (c) PK and dosing
         (1) Bioavailability 91% – Can be administered as a parenteral solution or oral tablets
         (2) Half-life: 12 hours
         (3) Dosing: 200 mg once daily

(d) Studies:
(1) Tedizolid once daily for 6 days was compared with linezolid twice daily for 10 days for the treatment of acute bacterial skin and skin structure infection and was found to be noninferior.
(2) Tedizolid once daily for 7 days was compared with linezolid twice daily for 10 days for the treatment of gram-positive hospital-acquired or ventilator-associated bacterial pneumonia and was found to be noninferior for day 28 all-cause mortality, but did not meet noninferiority for clinical response at test of cure.
(e) Application: Place in therapy among critically ill patients is limited, but tedizolid use may be warranted in patients that experience linezolid-associated adverse effects.

ii. Ceftaroline
(a) Antimicrobial class: Fifth-generation cephalosporin
(b) Spectrum of activity: Enterobacterales (similar to third-generation cephalosporins) and gram-positive pathogens, including MRSA, drug-resistant *S. pneumoniae*, and vancomycin-intermediate and vancomycin-resistant staphylococci
(c) PK and dosing
(1) Activity against MRSA mediated through enhanced affinity to penicillin-binding-protein 2a
(2) Half-life: 2.7 hours
(3) Excretion: 88% urine
(4) Dosing:
(A) 600 mg intravenously every 12 hours is the FDA label-approved dosing. Reports of using 600 mg intravenously every 8 hours for severe infections
(B) Adjustments necessary for patients with renal impairment
(d) Studies:
(1) Evaluated in a noninferiority trial compared with ceftriaxone, both in conjunction with clarithromycin, for the treatment of community-acquired bacterial pneumonia. Results indicated noninferiority, with numerically higher cure rates in those randomized to ceftaroline. Clinical response by day 4 of therapy was also higher in the ceftaroline group.
(2) Evaluated in a noninferiority trial compared with vancomycin plus aztreonam for the treatment of skin and skin structure infection. Ceftaroline cure rates were within the *a priori*–determined margin, thus satisfying the criteria for noninferiority.
(e) Application:
(1) FDA approved for acute bacterial skin and skin structure infections and community-acquired bacterial pneumonia
(2) Despite its FDA approvals, the main role of ceftaroline is the availability of an additional agent with activity against MRSA. May be an option when vancomycin therapy is suboptimal and other treatment options may lead to unwanted adverse effects. May also be a reasonable option for the treatment of community-acquired pneumonia in institutions where community-acquired MRSA is common and for the salvage treatment of MRSA bacteremia

iii. Ceftobiprole
(a) Cephalosporin antibiotic
(b) FDA approved in 2024 for the treatment of S. aureus bacteremias (including those with right-sided endocarditis), SSIs, and community-acquired pneumonia
(c) Has activity against MRSA

(d) Studies
(1) Showed noninferiority to daptomycin plus aztreonam for MRSA bacteremia
(2) Showed noninferiority to vancomycin plus aztreonam for SSIs
(3) Showed noninferiority to ceftriaxone plus optional linezolid for community-acquired pneumonia

Patient Case

6. A 72-year-old woman with a history of end-stage renal disease is admitted to the medical ICU with signs and symptoms of sepsis. Blood cultures are obtained in which two of two bottles grow gram-positive cocci. The patient is initiated on vancomycin, and her tunneled dialysis catheter is removed. On day 4 of therapy, the blood cultures are finalized to be MRSA with the following antibiotic susceptibility results: oxacillin greater than 4 mcg/mL – R; vancomycin 2 mcg/mL – S; daptomycin 0.5 mcg/mL – S; and linezolid 1 mcg/mL – S. A transthoracic echocardiogram is obtained, which reveals echodensities on the mitral valve. The patient will be treated medically with antibiotics for 4–6 weeks. The patient's repeat blood cultures are currently no growth. Her vital signs and laboratory values are as follows: blood pressure 150/90 mm Hg, heart rate 88 beats/minute, temperature 100.2°F, WBC 11 x 10^3 cells/mm^3, and lactate 1.1 mmol/L. Which is the most appropriate treatment regimen?
A. Change vancomycin to linezolid 600 mg every 12 hours.
B. Add gentamicin and rifampin to vancomycin.
C. Change vancomycin to daptomycin 6 mg/kg every 48 hours.
D. Continue vancomycin, and target a trough level of 15–20 mcg/mL.

VII. IMMUNOCOMPROMISED PATIENTS

A. Febrile Neutropenia
1. Definition
a. Fever: A single temperature of 101°F (38.3°C) or greater orally or a temperature of 100.4°F (38.0°C) or greater orally for more than 1 hour
b. Neutropenia: Less than 500 neutrophils/mm^3 or less than 500 neutrophils/mm^3 during the next 48 hours
2. Patients with neutropenic fever admitted to an ICU should be considered high risk.
3. Initial therapy should include monotherapy with an intravenous antipseudomonal β-lactam (i.e., cefepime, piperacillin/tazobactam, meropenem, imipenem). Patients with type 1 hypersensitivity should be treated with either ciprofloxacin or aztreonam plus vancomycin.
4. Consider dual gram-negative therapy (fluoroquinolone or aminoglycosides) in patients with shock or if antimicrobial resistance is suspected.
5. Consider adding vancomycin to gram-negative therapy in patients with shock, suspected catheter-related infection, skin and soft tissue infection, and/or pneumonia. Gram-positive therapy can be discontinued in 48–72 hours if no evidence of gram-positive infections is discovered.
6. Modifications to initial antibiotic choices should be considered for patients with worsening clinical status or if patients' microbiological data warrant change.
7. Unexplained persistent fever in an otherwise clinically stable patient rarely warrants an escalation in therapy. Persistent fevers for 4–7 days after initiation of antibacterial agents should warrant consideration for empiric antifungal coverage in those who have persistent neutropenia.

8. Initial antimicrobials should be de-escalated or escalated in documented infections depending on in vitro susceptibility. Documented infections and unexplained fevers should be treated for a minimum of 14 days and until the absolute neutrophil count is greater than 500 cells/mm^3 (whichever is later).
9. Patients with hemodynamic instability should have their initial antibiotic regimen escalated to include coverage for resistant bacteria and fungi.
10. Hematopoietic growth factors should not be used for the treatment of febrile neutropenia. Prophylactic use of hematopoietic growth factors should be considered for patients with a high anticipated risk of febrile neutropenia (20% or greater).

B. Solid Organ Transplantation
   1. Epidemiology
      a. Hospital-acquired bacterial infections are the most common types of infections in SOT recipients.
      b. 50%–75% of SOT recipients will experience an infection within the first year after transplantation.
      c. Posttransplant infections may contribute to graft dysfunction and reduce long-term survival, and they have been associated with prolonged LOS and cost of care.
      d. An analysis of 60,000 renal transplant recipients found that infections were the second leading cause of death.
   2. General risk factors for infections
      a. Excessive use of antibiotics before transplantation: With the use of prophylactic antimicrobials in the pretransplant period, many transplant recipients are experiencing resistant pathogens in the posttransplant period.
      b. Infections are most common in the first 6 months after transplantation, with different pathogens presenting after various durations of immunosuppression. See Figure 3. OIs are rare in the first month after transplantation because the full effects of immunosuppression are not yet present. Fungal and viral infections experienced during the first month after transplantation are usually donor derived.
      c. Duration of hospitalization after transplantation
      d. Renal dysfunction
      e. Several acute rejection episodes
   3. General clinical approach for infectious diseases issues in critically ill SOT recipients
      a. Be cognizant of the patient's transplant timeline, particularly with respect to possible OIs, pretransplant risk factors, immune status, intensity of immunosuppressive therapy, prophylactic regimens, and recent treatments of rejection.
      b. Have a high clinical suspicion for OIs.
      c. Severe infections sometimes warrant a reduction in maintenance immunosuppression. Consult with the transplant team regarding plans for maintenance immunosuppression.
      d. Be aware of the many drug interactions between immunosuppressants and antimicrobials.

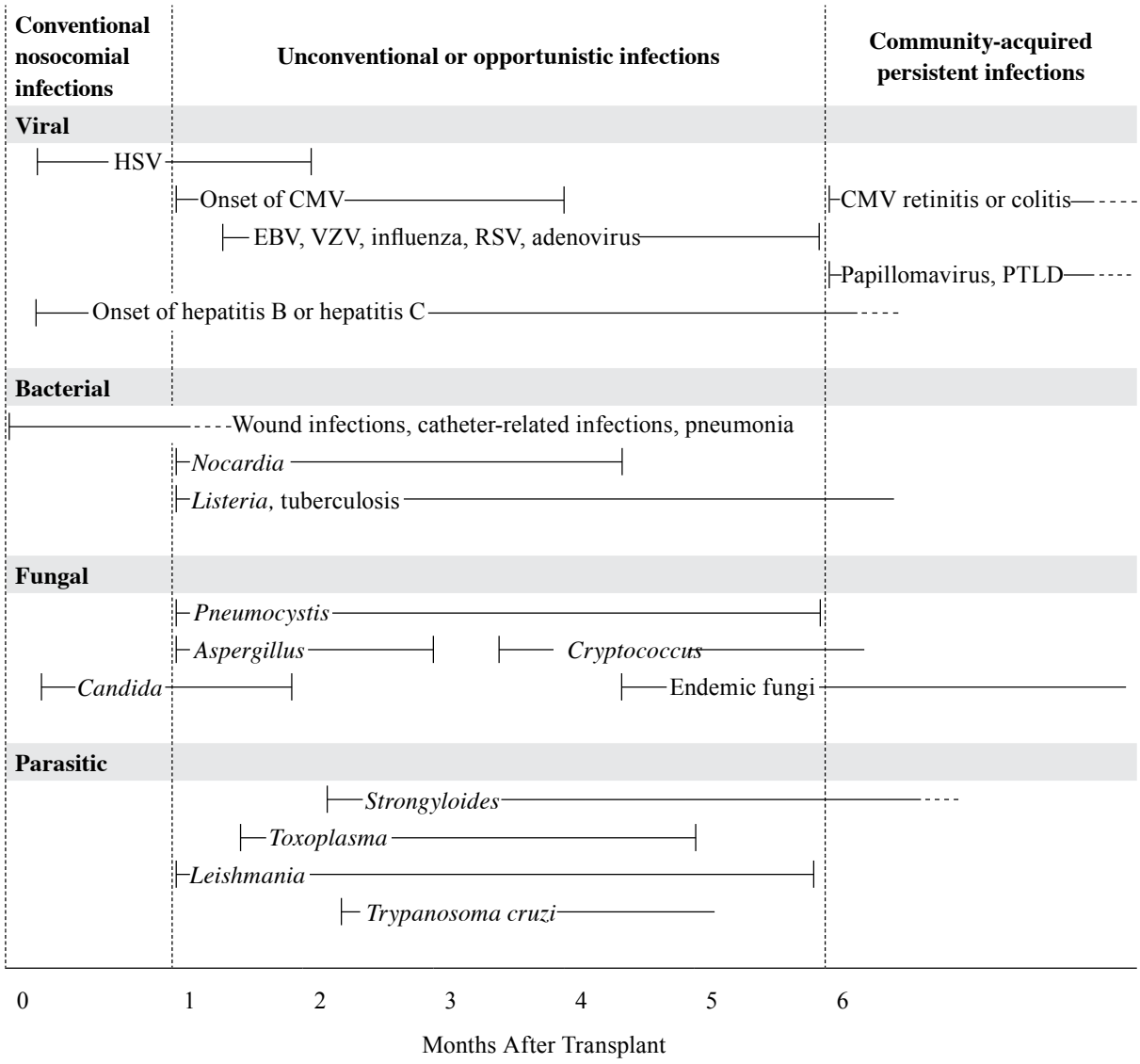


Figure 3. Usual sequence of infections after organ transplantation.[a]

[a]Zero indicates the time of transplantation. Solid lines indicate the most common period for the onset of infection; dotted lines indicate periods of continued risk at a reduced level.

CMV = cytomegalovirus; EBV = Epstein-Barr virus; HSV = herpes simplex virus; PTLD = posttransplant lymphoproliferative disease; RSV = respiratory syncytial virus; VZV = varicella-zoster virus.

Reproduced with permission from: Baillie GM. Infectious disease concerns in solid organ transplantation. In: Schumock GT, Brundage DM, Dunsworth TS, et al., eds. Pharmacotherapy Self-Assessment Program, 5th ed. Book 2. Transplantation. Lenexa, KS: American College of Clinical Pharmacy, 2004:165-86.

VIII. HUMAN IMMUNODEFICIENCY VIRUS/ACQUIRED IMMUNE DEFICIENCY SYNDROME IN CRITICALLY ILL PATIENTS

A. Diagnosis
   1. Isolated $CD4^+$ lymphopenia has occurred in critically ill patients without HIV disease.
      a. Differential diagnosis of $CD4^+$ lymphopenia
         i. Idiopathic
         ii. Common variable immunodeficiency
         iii. Corticosteroid administration
         iv. Circadian rhythm
         v. Hematologic malignancies
         vi. Critical illness
   2. Patients with traditional HIV risk factors, isolated $CD4^+$ lymphopenia, or AIDS-defining OIs who have not been given a diagnosis should receive HIV serologic testing. See Table 10 for AIDS-defining conditions.

B. Presentation for HIV-Associated Acute Illness
   1. Changing prescribing patterns in highly active antiretroviral therapy (HAART) and improvements in care have changed the landscape of ICU admissions associated with HIV. Although rates of ICU admissions among HIV-positive patients have remained stable over time, the rationale for ICU admission has shifted from respiratory distress associated with pneumocystic pneumonia to other sepsis etiologies, neurologic disorders, and other end-organ dysfunctions.
   2. See Table 11 for the management and prophylaxis of commonly occurring OIs in critically ill patients.

C. Management of HAART Therapy
   1. Treatment-naive patient
      a. Initiation of HAART in an ICU may be deterred by factors such as lack of enteral access, inability to assess the patient's willingness to commit to therapy, and high probability of interruptions in therapy because of surgical interventions.
      b. Immune reconstitution inflammatory syndrome
         i. Unexpected worsening of existing OI or unmasking of previously unrecognized illness associated with recent HAART initiation
         ii. Recovery of immune response leading to a proinflammatory cytokine storm
         iii. Risk factors
            (a) Presence of disseminated disease or high antigen load
            (b) High baseline HIV viral load
            (c) Rapid response to antiretroviral therapy
         iv. Evidence does not suggest that early administration of HAART (at the onset of an AIDS-defining OI) is associated with a higher incidence of immune reconstitution inflammatory syndrome.
         v. In general, early and continued administration of HAART associated with improved outcomes, particularly in patients with OI. For patients with severe symptoms associated with immune reconstitution inflammatory syndrome (requiring vasopressors or intubation) consider the adjunctive use of prednisone.
   2. Patient receiving HAART regimen
      a. Considerations: Availability of drug formulations conducive to ICU administration; food requirements; dose adjustments associated with renal or hepatic impairment; new drug interactions; possible interruptions in therapy

b. In general, if one of the antiretroviral therapies has to be discontinued, all of the therapies should be discontinued to decrease the promotion of resistance caused by the suboptimal suppression of viral replication. Rationale: Nonnucleoside reverse transcriptase inhibitors (NNRTIs) have long half-lives (as long as 3 weeks); therefore, if NNRTIs are discontinued at the same time as other antiretrovirals with shorter half-lives, there will be a period of functional NNRTI monotherapy. May consider continuing antiretrovirals with shorter half-lives, if possible, for 1–2 weeks to minimize selection of NNRTI resistance.
c. Possible ICU drug interactions – See Table 12.

3. HAART-associated adverse drug reactions
   a. Newer generations of HAART regimens are generally well tolerated; however, many patients still receive older HAART therapy.
   b. Many of the HAART-associated adverse effects have no specific treatment; therefore, early recognition and prompt discontinuation of the offending agent is crucial.
   c. Lactic acidosis
      i. Two nucleoside/nucleotide reverse transcriptase inhibitors (NRTIs) (e.g., lamivudine and tenofovir) remain the backbone of many HAART regimens.
      ii. NRTIs are associated with a variety of mitochondrial toxicities.
      iii. Older NRTIs, such as zidovudine, stavudine, and didanosine, have been associated with lactic acidosis.
      iv. Symptoms: Fatigue, malaise, nausea, vomiting, abdominal pain, hepatomegaly
      v. Management: Supportive therapy and discontinuing the potential offending agent
   d. Abacavir hypersensitivity
      i. Symptoms: Fever, rash, gastrointestinal (GI) symptoms
      ii. Reaction associated with the presence of the *HLA-B*5701* allele, which has an 8% prevalence among whites in North America. Genetic screening for this allele is recommended.
      iii. Management: Supportive care and discontinuing agent, with rechallenge contraindicated because of the possibility of life-threatening hemodynamic compromise
   e. Other HAART-associated significant adverse effects:
      i. Nevirapine hypersensitivity: Rash (may progress to Stevens-Johnson syndrome), fever, hepatotoxicity
      ii. Raltegravir-associated rhabdomyolysis
      iii. Tipranavir-associated hepatotoxicity and intracranial hemorrhage
      iv. Protease inhibitor–associated pancreatitis

D. Management and Prophylaxis of HIV-Associated Infections: See Table 10.

Table 10. AIDS-Defining Conditions

Bacterial infections, multiple/recurrent (children < 13 yr)	Lymphoid interstitial pneumonia or pulmonary lymphoid hyperplasia complex (children < 13 yr)
Candidiasis: Bronchi, trachea, or lungs	Lymphoma: Burkitt (or equivalent term)
Candidiasis: Esophageal	Lymphoma: Immunoblastic (or equivalent term)
Cervical cancer: Invasive (> 13 yr)	Lymphoma: Primary or brain
Coccidioidomycosis: Disseminated or extrapulmonary	*Mycobacterium avium* complex or *M. kansasii*: Disseminated or extrapulmonary
Cryptococcosis: Extrapulmonary	*M. tuberculosis*: Any site (pulmonary or extrapulmonary)
Cryptosporidiosis: Chronic intestinal (for > 1 mo)	*Mycobacterium*: Other species or unidentified species; disseminated or extrapulmonary
Cytomegalovirus disease (other than the liver, spleen, or nodes)	*P. jiroveci* pneumonia
Cytomegalovirus retinitis (with loss of vision)	Pneumonia: Recurrent
Encephalopathy: Related to HIV	Progressive multifocal leukoencephalopathy
Herpes simplex: Chronic ulcer(s) (for > 1 mo); bronchitis, pneumonitis, or esophagitis	*Salmonella septicemia* (recurrent)
Histoplasmosis: Disseminated or extrapulmonary	Toxoplasmosis of brain
Isosporiasis: Chronic intestinal	Wasting syndrome caused by HIV

Patient Case

7. A 46-year-old man is admitted to the medical ICU for diabetic ketoacidosis. The patient has a history of insulin-dependent diabetes, is HIV positive, and has cryptococcal meningitis. His current HAART regimen consists of atazanavir, ritonavir, tenofovir, and emtricitabine, which is continued on admission to the ICU. The patient's $CD4^+$ count is 85 cells/mm^3. The patient's diabetic ketoacidosis is well controlled, and he is ready to be discharged from the ICU. Before discharge, the patient is noted not to be on any prophylaxis against OIs. Which prophylactic regimen would be most appropriate for the patient?
   A. Azithromycin 1200 mg once weekly and trimethoprim/sulfamethoxazole 1 double-strength (DS) tablet thrice weekly.
   B. Fluconazole 200 mg daily and trimethoprim/sulfamethoxazole 1 DS tablet daily.
   C. Azithromycin 1200 mg once weekly and fluconazole 200 mg daily.
   D. Fluconazole 200 mg daily and trimethoprim/sulfamethoxazole 1 DS tablet thrice weekly.

Table 11. AIDS-Associated OIs and Their Respective Prophylactic and Treatment Options

OI/CD4+ Threshold for Prophylaxis	Prophylaxis Option	Treatment Option
P. jiroveci: < 200 cells/mm^3 and/or CD4 cell percentage < 14%)	Preferred: TMP/SMX[b] 1 DS PO daily *or* TMP/SMX[b] 1 SS PO daily Alternatives Dapsone[a] PO 100 mg/day Pentamidine 300 mg inhalation once monthly TMP/SMX 1 DS PO TIW Atovaquone PO 1500 mg/day	Preferred: TMP/SMX[b] 15–20 mg/kg PO/IV divided q6–8hr x 21 days *plus* prednisone 40 mg PO q12hr x 5 days; 40 mg/day x 5 days; 20 mg/day x 11 days (if Pao_2 < 70 mm Hg or alveolar-arterial O_2 gradient > 35) (IV methylprednisolone could be administered at 75% of the corresponding oral prednisone dose) Alternatives: Pentamidine[b] IV 4 mg/kg/day Clindamycin 600–900 mg PO/IV q6–8hr + primaquine[a] PO 30 mg/day (base) Atovaquone 750 mg PO q12hr – mild disease
Toxoplasmosis: < 100 cells/mm^3 and *Toxoplasma* IgG positive	Preferred: TMP/SMX 1 DS PO daily Alternatives: Dapsone[a] PO 50 mg/day + (pyrimethamine 50 mg PO + leucovorin 25 mg PO) once weekly TMP/SMX 1 DS PO TIW	Preferred: Pyrimethamine[b] 200 mg PO x 1; then 50–75 mg/day + sulfadiazine[b] 1000–1500 mg PO q6hr + leucovorin PO 10–20 mg/day x 6 wk Alternatives: Pyrimethamine[b] 200 mg PO x 1; then 50–75 mg/day + clindamycin 600–900 mg PO/IV q8hr + leucovorin PO 10–20 mg/day Pyrimethamine[b] 200 mg PO x 1; then 50–75 mg/day + atovaquone 1500 mg PO BID + leucovorin PO 10–20 mg/day
Mycobacterium avium complex: < 50 cells/mm^3	Preferred: Azithromycin 1200 mg once weekly *or* Clarithromycin 500 mg q12hr *or* Azithromycin 600 mg twice weekly Alternatives: Rifabutin[c] 300 mg/day	Preferred: At least two drugs: Clarithromycin 500 mg PO BID *or* azithromycin 600 mg/day *plus* ethambutol[b] PO 15–25 mg/kg/day Additional agents (consider depending on susceptibility testing, severe disease, or low CD4+ count) Rifabutin[c] PO 300 mg/day Amikacin[b] IV 10–15 mg/kg/day fluoroquinolones (moxifloxacin PO/IV 400 mg/day or levofloxacin[b] PO/IV 500 mg/day)

Table 11. AIDS-Associated OIs and Their Respective Prophylactic and Treatment Options *(continued)*

OI/CD4+ Threshold for Prophylaxis	Prophylaxis Option	Treatment Option
Mycobacterium tuberculosis: Primary prophylaxis not indicated	Latent infection treatment Preferred: Isoniazid (INH) PO 300 mg/day + pyridoxine PO 50 mg/day Alternatives: INH 900 mg PO twice weekly + pyridoxine 100 mg PO twice weekly	Active infection treatment Preferred: Rifampin[c] PO/IV 600 mg/day + INH PO 300 mg/day + pyrazinamide[b] PO 20–25 mg/kg/day + ethambutol[b] PO 15–25 mg/kg/day
Cryptococcosis: Primary prophylaxis not indicated Secondary prophylaxis may be considered < 100 cells/mm^3	Secondary prophylaxis: Fluconazole 200–400 mg/day	Preferred: Induction therapy x 2 wk: Liposomal amphotericin IV 3-4 mg/kg/day + flucytosine[b] PO 25 mg/kg q6hr Alternatives: Induction x 2 wk: 1. Amphotericin B IV 0.7–1 mg/kg/day + flucytosine[b] PO 25 mg/kg q6hr 2. Amphotericin B lipid complex 3-4 mg/kg/day + flucytosine[b] PO 25 mg/kg q6hr 3. Liposomal amphotericin IV 3–4 mg/kg/day + fluconazole[b] 800 mg PO/IV daily Preferred: Consolidation therapy x 8 wk Fluconazole PO/IV 400 mg/day Preferred: Maintenance therapy × 1 yr Fluconazole PO/IV 200 mg/day and 8 hr after infusion
Cytomegalovirus: Primary prophylaxis not indicated Secondary prophylaxis may be considered	Secondary prophylaxis: Valganciclovir[b] PO 900 mg/day	Preferred: Induction: Valganciclovir[b] PO 900 mg BID for 14–21 days *or* Ganciclovir[b] 5 mg/kg q12hr for 14–21 days Alternatives: 1. Foscarnet[b] IV 90 mg/kg q12hr or 60 mg/kg q8hr 2. Cidofovir IV 5 mg/kg/wk (with probenecid 2 g 3 hr before cidofovir; then 1 g 2 hr after cidofovir and 1 g 8 hr after the dose)

[a]Should not be used in patients with a glucose-6-phosphate dehydrogenase deficiency

[b]Renal adjustments may be necessary.

[c]Monitor for interactions with HAART.

BID = twice daily; DS = double strength; hr = hour(s); INH = isoniazid; IV = intravenous(ly); OI = opportunistic infection; PO = orally; SS = single strength; TIW = thrice weekly; TMP/SMX = trimethoprim/sulfamethoxazole.

Reproduced with permission from: Smith CL. HIV/infectious disease. In: ACCP Updates in Therapeutics® 2014: Pharmacotherapy Preparatory Review and Recertification Course. Lenexa, KS: American College of Clinical Pharmacy, 2014:580-621.

Table 12. Common Interactions Between Antiretrovirals and Commonly Used Medications in Critically Ill Patients

Agent	Antiretroviral	Interactions
Stress ulcer prophylaxis (proton pump inhibitors, histamine-2 receptor antagonists)	Rilpivirine: Contraindicated with proton pump inhibitors Atazanavir: Relatively contraindicated with proton pump inhibitors (administer no more than the equivalent of omeprazole 20 mg daily, and separate administration by at least 12 hr)	Decrease in antiretroviral concentrations
Triazole antifungals: Voriconazole, posaconazole, itraconazole, isavuconazole	Protease inhibitors Nonnucleoside reverse transcriptase inhibitors	Increase in antiretroviral concentration Decrease in antifungal concentrations
Antibacterials: Rifampin Clarithromycin Metronidazole	 Protease inhibitors Nonnucleoside reverse transcriptase inhibitors Protease inhibitors Nonnucleoside reverse transcriptase inhibitors Fosamprenavir, lopinavir, ritonavir	 Decrease in antiretroviral concentration Increase in clarithromycin concentration Disulfiram reaction
Antiarrhythmics: Amiodarone Flecainide, propafenone, quinidine Diltiazem	 Indinavir, ritonavir, tipranavir Lopinavir, ritonavir, tipranavir Atazanavir, fosamprenavir	 Increased antiarrhythmic concentrations
Statins	Protease inhibitors Nonnucleoside reverse transcriptase inhibitors	Increase concentrations of statins – in decreasing order of interaction potential (lovastatin, simvastatin, rosuvastatin, atorvastatin, pravastatin)
Anticonvulsants: Carbamazepine, phenobarbital, phenytoin	Protease inhibitors Nonnucleoside reverse transcriptase inhibitors	Decrease in antiretroviral concentrations Increase or decrease in anticonvulsant concentrations
Midazolam	Protease inhibitors Nonnucleoside reverse transcriptase inhibitors Cobicistat/elvitegravir	Increase in midazolam concentration
Methadone	Nonnucleoside reverse transcriptase inhibitors; fosamprenavir, ritonavir, lopinavir, nelfinavir, didanosine, saquinavir	Decrease in methadone concentration (potentially leading to opioid withdrawal)

Table 12. Common Interactions Between Antiretrovirals and Commonly Used Medications in Critically Ill Patients *(continued)*

Agent	Antiretroviral	Interactions
Sildenafil	Protease inhibitors; delavirdine	Increase in sildenafil concentration
Anticoagulants: Warfarin Apixaban Rivaroxaban	 Delavirdine, efavirenz Protease inhibitor/cobicistat, protease inhibitor/ritonavir Protease inhibitor/cobicistat, protease inhibitor/ritonavir	Increase in warfarin concentration Increases in apixaban concentrations: Do not administer in patients who require 2.5 mg BID; in patients requiring 10 mg or 5 mg twice daily, reduce apixaban by 50% Increases in rivaroxaban expected: Do not coadminister

IX. ANTIFUNGAL THERAPY

A. Amphotericin B
   1. Mechanism of action: Binds to ergosterol in the fungal cell membrane, which alters the membrane permeability, leading to ion leakage and fungal cell death
   2. Spectrum of activity
      a. *Candida* spp. (except for *C. lusitaniae*), Blastomycetes, coccidioidomycosis, *Cryptococcus*, *Paracoccidioides*, histoplasmosis, aspergillosis, mucormycosis
      b. Wide spectrum of activity allows its clinical use in many different systemic fungal infections.
   3. Dose
      a. Conventional amphotericin: 0.5–1.0 mg/kg/day
      b. Lipid amphotericin: 3–5 mg/kg/day (higher doses have not been associated with improved outcomes)
   4. Adverse effects
      a. Renal toxicity
         i. Decrease in glomerular filtration rate, which is associated with cumulative doses greater than 4–5 g
         ii. Clinical manifestation
            (a) Renal tubular acidosis
            (b) Oliguria
            (c) Azotemia
            (d) Potassium, magnesium, phosphate wasting
         iii. Prevention
            (a) Avoid concomitant nephrotoxins.
            (b) Avoid dehydration.
            (c) Salt loading: 500 mL of normal saline before and after infusion of amphotericin
      b. Infusion-related reactions
         i. Mediated by cytokine release and prostaglandin synthesis
         ii. Presentation: Fever, chills, nausea, vomiting, flushing, rigors
         iii. Prevention

(a) Premedications – Administered 30–60 minutes before infusion
(1) Hydrocortisone 25–50 mg intravenously/orally
(2) Ibuprofen 600 mg orally
(3) Acetaminophen 650 mg orally with diphenhydramine 50 mg orally
(4) Meperidine 25–50 mg intravenously if patient previously experienced rigors
(b) Use lower infusion rate on dose initiation.

c. Lipid amphotericin (liposomal amphotericin and amphotericin B lipid complex)
i. Amphotericin dissociates from lipid over time, which may limit its renal toxicity and infusion-related reactions.
ii. Premedication may still be necessary for amphotericin B lipid complex.

B. Triazole Antifungals
1. Mechanism of action: Inhibits the synthesis of ergosterol through blocking the CYP enzyme 14-α-sterol-demethylase. Inhibition of this enzyme leads to the accumulation of 14-α-methyl sterols on the fungal surface, which in turn leads to fungal cell death.
2. Fluconazole
a. Spectrum of activity:
i. *Candida* spp.
(a) *Candida krusei* is intrinsically resistant.
(b) Variable sensitivity with *Candida glabrata*. Should not be used for *C. glabrata* unless antifungal susceptibilities are available. If *C. glabrata* is sensitive dose-dependent, higher doses may be necessary (12 mg/kg/day).
(c) Good activity against all other pathogenic *Candida* spp.
ii. *Cryptococcus*
b. Dose
i. 6–12 mg/kg/day (400–800 mg/day)
ii. Available in intravenous and oral formulations
iii. Well absorbed orally with high bioavailability
iv. Primarily excreted by the kidneys – Renal dose adjustments are necessary.
c. Adverse effects: Minimal adverse effects and lowest propensity for drug interactions among triazoles
3. Itraconazole
a. Spectrum of activity
i. *Candida* spp. – Has coverage similar to fluconazole with the addition of activity against *C. krusei*
ii. *Aspergillus*, *Blastomyces*, *Histoplasma*
b. Dose
i. 200–400 mg once daily
ii. Intravenous formulation no longer available
iii. Poor and erratic oral absorption. Improved with oral liquid formulation, maintaining high stomach acidity, avoidance of acid-suppressive therapy, and coadministration with acidic beverage
iv. Therapeutic drug monitoring may be necessary. Target: Itraconazole plus hydroxy-itraconazole concentration greater than 1 mcg/mL
v. Primarily used for fungal prophylaxis in immunocompromised patients and treatment of endemic fungi (histoplasmosis, blastomycosis)

c. Adverse effects
    i. GI upset, increase in liver function tests
    ii. Drug interactions: CYP 3A4 and 2C9 inhibitor

4. Voriconazole
    a. Spectrum of activity
        i. *Candida* spp. – Has coverage similar to fluconazole with the addition of activity against *C. krusei*
        ii. *Aspergillus*, *Fusarium*, *Scedosporium*: Resistance against voriconazole has occurred with these pathogens.
    b. Dose
        i. Drug of choice for invasive aspergillosis
        ii. 6 mg/kg every 12 hours x 2 doses as the loading dose, followed by 4 mg/kg every 12 hours
        iii. Intravenous and oral formulations are available.
        iv. Intravenously formulated in sulfobutyl-ether-β-cyclodextrin, which accumulates in renal dysfunction, although the clinical significance of this is unknown
        v. Extensively metabolized by the liver, with 50% dose reductions recommended for patients with moderate to severe cirrhosis
        vi. Genetic variations in CYP metabolism and high propensity for drug interactions lead to wide interpatient variability in concentrations.
        vii. Therapeutic drug monitoring may be necessary. Target trough concentration: 1–5.5 mcg/mL.
    c. Adverse reactions
        i. Increase in liver function tests
        ii. Visual hallucinations
        iii. Rash
        iv. Nausea
        v. CYP 3A4 and 2C9 inhibitor:
            (a) Contraindicated with the use of rifampin, rifabutin, sirolimus, barbiturates, carbamazepine, and quinidine
            (b) Significant dose reductions for cyclosporine and tacrolimus when coadministered with voriconazole
        vi. QTc prolongation

5. Posaconazole
    a. Spectrum of activity: Wide spectrum of activity, which includes *Candida* (similar to voriconazole), *Aspergillus*, *Zygomycetes*, and *Fusarium*
    b. Dose
        i. Oral suspension (immediate release): 200 mg every 6 hours. Oral suspension has extremely erratic absorption that is enhanced by coadministration with a high-fat meal and acidic food. In critically ill patients in whom coadministration with fatty meals is not possible and avoidance of acid-suppressive stress ulcer prophylaxis cannot be avoided, would recommend considering an alternative therapy or administration method (e.g., oral tablets or intravenous)
        ii. Oral tablets (delayed release): 300 mg every 12 hours x 2 doses, followed by 300 mg once daily. Oral tablet is in an extended-release formulation, which cannot be crushed. Oral tablet absorption not as dependent on gastric pH and meal lipid content.
        iii. Intravenous: 300 mg every 12 hours x 2 doses, followed by 300 mg once daily (intravenously formulated in sulfobutyl-ether-β-cyclodextrin, which accumulates in renal dysfunction, although the clinical significance of this is unknown)
        iv. Therapeutic drug monitoring may be prudent, particularly when oral suspension is used. Target concentrations: Trough greater than 1.25 mcg/mL for treatment of invasive fungal infection

v. Mainly used for fungal prophylaxis in immunocompromised patients and treatment when patient is not responding to other therapies

c. Adverse reactions
   i. Increase in liver function tests
   ii. Nausea/vomiting
   iii. Drug interaction: CYP3A4 inhibitor
   iv. QTc prolongation

6. Isavuconazole
   a. Spectrum of activity: Wide spectrum of activity, including *Candida* (similar to voriconazole), *Aspergillus* (may retain activity for some species that are resistant to other azoles), *Zygomycetes*, and dimorphic fungi. Limited activity against *Fusarium*.
   b. Dose
      i. Isavuconazonium: Loading dose 372 mg (equivalent to 200 mg of isavuconazole) every 8 hours intravenously × 6 doses, maintenance dose 372 mg (equivalent to 200 mg of isavuconazole) daily
      ii. Isavuconazonium sulfate solution is readily water soluble, unlike posaconazole and voriconazole, and does not require stabilization with cyclodextrin.
   c. Mainly PK advantages and safety benefits over voriconazole and posaconazole
      i. Linear kinetics
      ii. Oral bioavailability: 98%, not affected by food or acidity
      iii. Fewer drug-drug interactions
      iv. No QTc prolongation; in fact, associated with dose-dependent QTc shortening

C. Echinocandins (caspofungin, micafungin, rezafungin, and anidulafungin)
   1. Mechanism of action: Inhibition of glucan synthase, which is an enzyme responsible for the formation of 1,3-β-d-glucan. Inhibition of this enzyme leads to cessation of fungal cell wall formation.
   2. Spectrum of activity: All *Candida* spp. and *Aspergillus*
      a. In vitro, *C. parapsilosis* has a much higher MIC than other *Candida* spp. The clinical significance of this is unknown. Cases of breakthrough infections with *C. parapsilosis* during treatment with echinocandins have been reported. Several retrospective studies did not find worsened outcomes associated with echinocandin treatment of *C. parapsilosis*. The current IDSA guidelines on invasive candidiasis recommend using fluconazole for the treatment of *C. parapsilosis*.
      b. Primarily used for invasive candidiasis, neutropenic fever, and invasive *Aspergillus* if patient cannot tolerate other therapies. Recent IDSA candidiasis guidelines recommend echinocandin as preferred initial therapy for both proven and empiric therapy.
   3. Dose (available only as an intravenous formulation)
      a. Caspofungin: 70 mg once, followed by 50 mg once daily
      b. Micafungin: 100 mg once daily
      c. Anidulafungin: 200 mg once, followed by 100 mg once daily
      d. Rezafungin 400 mg intravenously once on day 1, followed by 200 mg intravenously weekly beginning on day 8 for up to four doses
   4. Adverse reactions
      a. Well tolerated with minimal GI adverse effects
      b. Minimal drug interactions
         i. Avoid caspofungin coadministration with cyclosporine and tacrolimus.
         ii. Avoid micafungin coadministration with nifedipine and sirolimus.
         iii. Avoid administering anidulafungin with metronidazole (disulfiram reaction).

5. Empiric treatment of critically ill patients: A recent large, multicenter study evaluated the use of empiric micafungin in immunocompromised critically ill patients with sepsis, multiple *Candida* colonization, multiorgan failure, and exposure to broad-spectrum antibiotics. The study showed no improvement in fungal infection-free survival compared with placebo. Until further evidence is available, empiric echinocandins should not be administered routinely.

D. Flucytosine
   1. Mechanism of action: Converted by fungal enzymes to fluorouracil, which disrupts fungal RNA and DNA synthesis
   2. Spectrum of activity
      a. *Candida* spp.
      b. *Cryptococcus*: Treatment of choice (in conjunction with amphotericin) for C*ryptococcus* meningitis
   3. Dose
      a. 25 mg/kg every 6 hours
      b. Renal adjustments necessary
      c. Well absorbed: Bioavailability 80%–90%
      d. Available only as an oral formulation in the United States
      e. Therapeutic concentrations: 25–100 mcg/mL
   4. Adverse reactions: Bone marrow suppression, particularly with supratherapeutic concentrations

Patient Case

8. A 66-year-old woman (height 66 inches, weight 75 kg) is admitted to the medical ICU for dehydration and acute kidney injury. The patient recently received an allogeneic bone marrow transplant and has not yet engrafted. She has been pancytopenic for 12 days. On day 5 of the medical ICU stay, the patient develops acute respiratory distress requiring intubation. Bronchoalveolar lavage is done, which eventually grows *Aspergillus fumigatus*, and the patient is given a diagnosis of invasive pulmonary aspergillosis. The patient's current medications include tacrolimus, corticosteroids, and fluconazole fungal prophylaxis. Her current relevant laboratory values are as follows: WBC 0.2 x 10^3 cells/mm^3, lactate 1.5 mmol/L, and SCr 3.4 mg/dL. Which antifungal therapy is most appropriate?
   A. Amphotericin 50 mg intravenously once daily.
   B. Isavuconazole load followed by 200 mg intravenously daily.
   C. Caspofungin load followed by 50 mg intravenously daily.
   D. Voriconazole load followed by 300 mg intravenously every 12 hours.

X. SOCIAL DETERMINANTS OF HEALTH AND INFECTIOUS DISEASES

A. Social determinants of health are categorized by the CDC into the following domains:
   1. Healthcare access and quality
   2. Economic stability
   3. Neighborhood and built environment
   4. Education access and quality
   5. Community and social context

B. Social determinants of health are a key focus of Healthy People 2030: "Create social, physical, and economic environments that promote attaining the full potential for health and well-being for all."

C. The WHO has put forth a conceptual framework for acting on social determinants of health to reduce health inequities and improve overall health and well-being.

D. Research has connected several social determinants of health factors with the following:
   1. Vaccination
   2. HIV
   3. Viral hepatitis
   4. Sexually transmitted diseases
   5. Tuberculosis
   6. COVID-19
      a. Many social determinants of health including poverty, physical environment, and race or ethnicity can have a considerable effect on COVID 19 outcomes.

E. Often people in poverty have limited access to health care, must work when they are sick, and face more stress, which can impact immunity and increase susceptibility to infection and death.

F. By remaining intentional about decreasing unconscious biases and seeing past medications to the individual to identify potential barriers, pharmacists can bolster the movement toward health equity.

G. The Pharmacy Quality Alliance issued a resource guide to help pharmacists navigate social determinants of health.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: D**

The CDC definitions for CAUTIs, which are used by the CMS reporting system, were recently updated. Because this patient had his urinary catheter removed for more than 24 hours before developing UTI symptoms, this case does not satisfy the definition of CAUTI (Answer D is correct; Answers A and C are incorrect). The current CDC CAUTI definitions do not consider the results of a urinalysis (Answer B is incorrect).

2. **Answer: A**

This patient presents with community-acquired meningitis. The high opening pressure is suggestive of a bacterial etiology, of which the most prevalent pathogen for this patient's risk factors and age is *S. pneumoniae*. Administering dexamethasone before initiating antibiotic therapy decreases mortality in patients with *S. pneumoniae* meningitis. (Answer A is correct) Awaiting laboratory and microbiologic test results analysis may cause inappropriate delays in therapy (Answer B is incorrect). In addition, although administering antimicrobials is a good idea, the beneficial effects associated with dexamethasone are diminished if administered after antimicrobials have been initiated. At the very least, corticosteroids should be administered concomitantly with antimicrobials (Answer C and D are incorrect).

3. **Answer: D**

The PNA FISH technology is designed to identify pathogens earlier. It performs well, with good sensitivity and specificity; hence, identification of *C. parapsilosis* should not require verification from culture (Answer A is incorrect). The drug of choice for the treatment of *C. parapsilosis* is fluconazole (Answer D is correct). Although echinocandin, voriconazole, and amphotericin would all cover *C. parapsilosis,* they all have a broader spectrum than is necessary, making them not the ideal choice (Answers A–C are incorrect).

4. **Answer: A**

Although PCT is a promising biomarker for the detection of bacterial infections, it must be interpreted within the context of the patient's clinical course. In this case, the patient has clear signs and symptoms of infection, together with signs of hypoperfusion. A low PCT should not be used to guide therapy discontinuation in this case, and antibiotics should be continued to cover the most likely pathogens (Answer A is correct; Answer B is incorrect). Because the patient is admitted with presumed community-acquired pneumonia, the combination of ceftriaxone and azithromycin is appropriate right now (Answers C and D are incorrect).

5. **Answer: B**

This patient is being treated for an ESBL *E. coli* UTI, in which the source of the infection (urinary catheter) is removed. The patient also has signs of response to piperacillin/tazobactam with resolution of leukocytosis and fever. In this case, given the low inoculum of the infection, initial response, removal of source, and low MIC, continuing piperacillin/tazobactam is the best choice (Answer B is correct). This choice may only be the case for UTIs, not other sources of infection (e.g., bacteremias). Traditionally, it was widely believed that ESBL infections had to be treated with carbapenems. However, recent evidence suggests that β-lactam/β-lactamase inhibitor combinations are a suitable option. In the era of antimicrobial stewardship, the preservation of carbapenem therapy should be regarded with high importance (Answer A is incorrect). This current culture has an MIC of 4 mcg/mL to cefepime, which, according to the updated CLSI guidelines, would be considered resistant (Answer C is incorrect). Because the patient is responding to the current therapy, combination therapy is unnecessary (Answer D is incorrect).

6. **Answer: D**

This patient is being treated for MRSA right-sided endocarditis, with the presumed source from her tunneled dialysis catheter. According to the IDSA guidelines for the treatment of MRSA, treatment with vancomycin of an isolate with an MIC of 2 mcg/mL or less should be determined by the patient's response to therapy, regardless of the actual MIC. This patient has responded to therapy with the presumed clearance of blood cultures and resolution of signs and symptoms of infection. Hence, continuing vancomycin and targeting a trough of 15–20 mcg/mL is the most appropriate choice (Answer D is correct). Linezolid is not indicated for the treatment of endocarditis (Answer A is incorrect). Adding gentamicin and rifampin is

considered in the medical treatment of prosthetic valve endocarditis (Answer B is incorrect). Because the patient is responding to current vancomycin therapy, an escalation to daptomycin is inappropriate right now (Answer C is incorrect).

7. **Answer: B**

This patient is admitted to the ICU for an indication that is unrelated to the patient's underlying HIV. The assessment of which prophylactics are necessary against OIs depends on the patient's underlying disease, history, and $CD4^+$ count. In this case, the patient has a $CD4^+$ count less than 100/mm^3 and a history of cryptococcal meningitis. Hence, prophylaxis should be administered for toxoplasmosis, *P. jiroveci*, and *Cryptococcus*. The best regimen for this patient is fluconazole 200 mg daily and trimethoprim/sulfamethoxazole 1 DS tablet once daily (Answer B is correct). The prophylaxis for *Mycobacterium avium* complex is indicated for patients with a $CD4^+$ count of less than 50 cells/mm^3, which is unnecessary at this point (Answers A and C are incorrect). Although trimethoprim/sulfamethoxazole 1 DS tablet thrice weekly is an option for prophylaxis, it is not the best choice against toxoplasmosis (Answer D is incorrect).

8. **Answer: B**

This patient has invasive pulmonary aspergillosis. Usually, the treatment of choice is voriconazole. However, in this case, the patient has acute kidney injury with a CrCl less than 50 mL/minute/1.73 m^2. According to the package insert, voriconazole is contraindicated in this case because of the possibility of accumulation of cyclodextrin, the intravenous drug carrier for voriconazole. Although the clinical relevance of this accumulation is controversial, continued use of a contraindicated therapy is inappropriate when alternatives may be available (Answer D is incorrect). Isavuconazole is a new triazole that was found to be noninferior to voriconazole for the treatment of aspergillosis and has improved water solubility, which does not require it to be formulated with cyclodextrin (Answer B is correct). Conventional amphotericin may be a reasonable choice, but it will likely worsen the patient's acute kidney injury (Answer A is incorrect). Echinocandins are not the ideal therapies for invasive aspergillosis and should only be considered if there are no other treatment options. In addition, caspofungin can decrease the serum concentration of tacrolimus (Answer C is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**

The prophylactic regimen for a patient without a β-lactam allergy who is undergoing cardiac surgery should be cephalosporin administered within 60 minutes of incision time, administered every 4 hours during surgery, and continued for no more than 48 hours. In this case, the patient has obesity disorder and weighs more than 120 kg; hence, the patient requires an initial dose of cefazolin 3 g (Answer C is correct). Given the patient's weight, administering cefazolin 2 g initially is inappropriate (Answers A and D are incorrect). Vancomycin is usually reserved for patients with a β-lactam allergy (Answer B is incorrect).

2. **Answer: C**

Although critical care pharmacists may not officially be part of many antimicrobial stewardship teams, many of their daily clinical activities constitute antimicrobial stewardship activities. These may include selecting the most appropriate treatment regimen and advocating the early de-escalation of antimicrobials. Even in the presence of a formalized antimicrobial stewardship team, these activities are often complementary to the formalized activities of the team (Answer C is correct). Given the wide variations in clinical practice, it may not be feasible to include an infectious diseases–trained pharmacist with every stewardship team. In that case, the activities and involvement of a critical care pharmacist may be even more crucial (Answer A is incorrect). Antimicrobial cycling has not consistently demonstrated beneficial effect on antimicrobial resistance. (Answer B is incorrect). Studies have shown that antimicrobial stewardship efforts in critically ill patients do not worsen outcomes. Given the aggressive empiric antimicrobial regimens commonly used in critically ill patients, antimicrobial stewardship should be instituted to minimize adverse effects and the emergence of resistance (Answer D is incorrect).

3. **Answer: D**

This patient presents with a health care–associated CNS infection, given the post-neurosurgical and device-related etiology of the infection. The most common pathogens include MRSA and multidrug-resistant gram-negative organisms. In addition to neurosurgical management of the device (e.g., removal or revision), empiric antibiotic therapy is indicated, including therapy with agents having empiric activity against suspected pathogens and the ability to safely achieve relevant CSF concentrations. Cefepime and vancomycin are the most appropriate options listed with consideration of CNS-specific dosing (Answer D is correct). Ceftriaxone does not cover nosocomial-acquired gram negatives such as *Pseudomonas* (Answers A and B are incorrect). Piperacillin/tazobactam would be reasonable according to the spectrum of antibacterial activity; however, poor CNS penetration of tazobactam limits the utility of this agent for meningitis and other CNS infections (Answer C is incorrect).

4. **Answer: D**

The presence of the CTX-M gene detected on *E. coli* by rapid diagnostic testing usually signifies the presence of ESBL. This may be why the patient has not yet responded to piperacillin/tazobactam. The most appropriate action at this point is to broaden the coverage to cover for potential ESBL-producing *E. coli*. Carbapenems remain the drug of choice for ESBL-producing organisms, particularly in a patient who is hemodynamically unstable (Answer D is correct). Extended-spectrum β-lactamases are usually encoded on genes that carry resistance against other classes of antimicrobials; hence, resistance to other antimicrobials is common; therefore, adding either aminoglycosides or fluoroquinolones may not be appropriate (Answers B and C are incorrect). Although at times ESBLs may be covered by cefepime, it must be determined by final AST (Answer A is incorrect).

5. **Answer: C**

E. cloacae are AmpC β-lactamase–producing Enterobacterales. The use of ceftriaxone or extended-spectrum penicillins (e.g., piperacillin and ticarcillin) may select out derepressed mutants, which are capable of causing the hyperproduction of AmpC β-lactamases. Derepressed mutants are capable of producing resistance against third-generation cephalosporins, monobactams, and extended-spectrum penicillins. In this case, the patient was taking 10 days of ceftriaxone before a new blood culture was growing lactose-positive gram-negative bacilli. Because lactose-positive gram-negative bacilli are usually Enterobacterales, growing multidrug-resistant pathogens such as *P. aeruginosa*

and *Acinetobacter baumannii* is less likely. Hence, the most likely resistance mechanism in this patient is either selection of derepressed mutants or acquisition of a pathogen with ESBL. Both of these resistance mechanisms are adequately treated by a carbapenem. Hence, changing to a carbapenem pending final sensitivities is the most reasonable option (Answer C is correct). Both types of resistance mechanisms are capable of producing resistance against ceftazidime and piperacillin/tazobactam (Answers A and B are incorrect). Because the patient developed a new bacteremia while taking ceftriaxone, it is not reasonable to continue ceftriaxone alone (Answer D is incorrect).

6. **Answer: B**

The patient is only on day 4 of therapy from proven MRSA pneumonia. However, the patient developed a bacteremia with gram-positive cocci in pairs and chains despite receiving systemic vancomycin therapy. The most likely culprit is a vancomycin-resistant *Enterococcus* sp. The medical team has already discontinued vancomycin; therefore, the new therapy must cover both the MRSA pneumonia and the possibility of a vancomycin-resistant *Enterococcus* sp. Linezolid has good lung penetration and can adequately cover vancomycin-resistant enterococci (Answer B is correct). Daptomycin would provide adequate coverage for vancomycin-resistant enterococci, but because it is inactivated by lung surfactants, it is not a good option for the treatment of MRSA pneumonia (Answer A is incorrect). Ceftaroline covers MRSA and has good lung penetration; however, it covers only vancomycin-resistant *E. faecalis*, not *E. faecium*. Because, at this point, the speciation of the gram-positive cocci in pairs and chains is not available, ceftaroline is not the best choice (Answer C is incorrect). Tigecycline does cover MRSA and vancomycin-resistant enterococci; however, given its large volume of distribution and relatively low serum concentrations, it is not the ideal choice for the treatment of bacteremia when other treatments are available (Answer D is incorrect).

7. **Answer: B**

The patient's history and clinical presentation suggest *Pneumocystis jiroveci* pneumonia. It is severe enough to warrant intubation, and the patient has a significant alveolar-arterial oxygen gradient. The usual drug of choice for such patients is trimethoprim/sulfamethoxazole, but because this patient has a sulfa allergy, this is not an option (Answer A is incorrect). According to the HIV OI guideline, the second-line agent for the treatment of severe *Pneumocystis jiroveci* pneumonia is intravenous pentamidine. Because the patient had severe hypoxemia, adjunctive steroids should be administered (Answer B is correct). Atovaquone and primaquine/clindamycin regimens are usually reserved for patients with milder *Pneumocystis jiroveci* pneumonia. Furthermore, primaquine should not be administered to someone with a glucose-6-phosphate dehydrogenase deficiency (Answers C and D are incorrect).

8. **Answer: B**

This patient has febrile neutropenia with no recovery of neutrophils. According to the IDSA febrile neutropenia guidelines, when a source of infection is identified, the empiric antimicrobial therapy can be de-escalated to a more narrow-spectrum regimen according to the antibiotic susceptibility report. In this case, because the *E. coli* was pan-sensitive, it would be appropriate to de-escalate to a narrow-spectrum antimicrobial. The guidelines also specify that antimicrobial therapies should be continued for at least 14 days and until neutrophils are greater than 500 cells/mm^3 (Answer B is correct). Although the patient continues to be febrile, an otherwise stable patient with continued fevers rarely requires additional antimicrobial therapy according to the guidelines. Hence, continuing more broad-spectrum therapy than necessary or adding other antimicrobials is unwarranted (Answers A, C, and D are incorrect).